

Unit 9 Overview: Data Sets

Unit Narrative



In this unit, students will discover how to classify, display, and interpret data. Students will summarize data from given displays using measures of center and variation. Measures of center and variation include mean, median, mode, range, and interquartile range. Displays include box plots, histograms, and line plots.

In Section 1, students will explore the difference between numerical and categorical data. They will also learn how to identify and create statistical questions that generate data. Knowing the characteristics of a statistical question will help foster understanding of data in the following sections.

In Section 2, students find and interpret the mean, median, mode, and range for given sets of data. This section builds on the understanding of finding mean, median, mode, and range from 5th grade and focuses on interpreting real-world data. Students will interpret data presented in box plots, histograms, and line plots. This section builds on what students have learned in 5th grade using tables and line plots. This course is the first time students will be interpreting box plots and histograms.

Section 3 focuses on creating data displays. Students build foundational understanding of what information is displayed in box plots and histograms in the previous sections before learning how to create the displays given a set of data.

This unit sets the foundation for Unit 10, Representing Data and Distributions, where students will dive deeper into data displays, outliers, comparing more than one data distribution, and using data distributions from a sample to make predictions about a population.

Unit At-A-Glance

Section 1: Types of Data	Benchmarks Addressed	Lesson 1	Types of Data
	MA.6.DP.1.1	Lesson 2	Statistical Questions
Section 2: Summarizing Data Distributions	Benchmarks Addressed	Lesson 3	Finding and Interpreting the Mean, Median, Mode, and Range
	MA.6.DP.1.2 MA.6.DP.1.3 MA.6.DP.1.4	Lesson 4	Interpreting Box Plots
		Lesson 5	Describing and Interpreting Histograms - Part 1
		Lesson 6	Describing and Interpreting Histograms - Part 2
		Lesson 7	Describing and Interpreting Line Plots
		Lesson 8	Comparing Distributions
Section 3: Creating Data Displays	Benchmarks Addressed	Lesson 9	Creating Box Plots
	MA.6.DP.1.3 MA.6.DP.1.5	Lesson 10	Creating Histograms



Differentiation

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Engagement <i>The “WHY” of learning</i>	• In lessons that collect student data, consider having students input the data into an Excel or Powerpoint document to create charts and graphs of the data to be presented. • Consider allowing students to input data into an Excel spreadsheet to create graphs and charts. Students can use the equation features to manipulate data to find the different features of a data set.
Representation <i>The “WHAT” of learning</i>	• Consider allowing students to use small whiteboards and dry erase markers to create charts and graphs to allow students to revise their work before answering in the workbook. • For the lessons that require student work to be displayed, consider leaving these pieces up throughout the week so that students can refer back to them as they progress through the unit.
Action & Expression <i>The “HOW” of learning</i>	• Provide graphic organizers for students when working with mean, median, mode, and range to help establish the differences between each category. • Consider providing a labeled template for each type of chart when students are first learning how to graph points. • Have students make a checklist with each learning target for students to check off after mastering each lesson to track progress.

ELL Information

Sentence Frames offer guided scaffolding for English Language Learners developing their skills in language acquisition and math literacy. Throughout this unit, students make sense of new concepts and terms through partner and small group discussions. Students reason about statistical questions and methodology through mathematical discourse. Provide sentence frames for English Language Learners as scaffolding for constructing sentences and explanations in the following lessons:

- Lesson 1: Types of Data
- Lesson 2: Statistical Questions
- Lesson 4: Interpreting Box Plots

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support.”

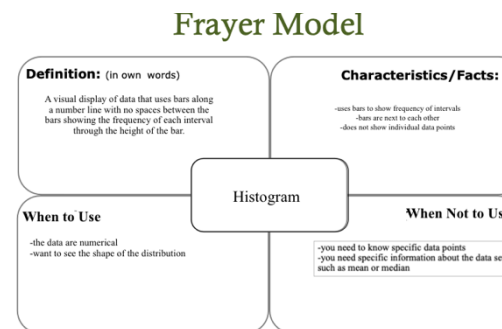
- | | |
|--|---|
| <ul style="list-style-type: none"> • Box plot • Cluster (data) • Frequency table • Interquartile range (IQR) • Mean • Measures of center • Mode • Quartile • Statistical question | <ul style="list-style-type: none"> • Categorical data • Data • Histogram • Line plot • Median • Measures of variability • Outlier • Range (of a data set) |
|--|---|

Additional Differentiated Support

Scaffolding Resources

Frayer Models are helpful graphic organizers because they allow students to see terms in multiple ways. This unit introduces students to many words that set the foundation for statistics.

- Provide the definition, but have students work in pairs to illustrate the word (when possible, or else list characteristics).
- Instead of examples and non-examples, have students record when to use and when not to use the different graphical representations, like line plots and histograms. They will build on this knowledge throughout the unit, so encourage students to see this as a work in progress.
- Consider creating a Frayer Model for each term from the Glossary by Section lists (above).



On-Ramp

On-Ramp to 6th grade provides video and practice tools for students to scaffold the following skills:

- Creating and Interpreting Line Plots
- Creating Line Plots with Fractions
- Using Line Plots to Solve Problems



Section Overview

Section 1: Types of Data (Lessons 1-2)

Benchmarks Addressed	Learning Targets	
MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.	Lesson 1	I can classify data as categorical or numerical.
	Lesson 2	I can write statistical questions that generate numerical data.

Section 2: Summarizing Data Distributions (Lessons 3-8)

Benchmarks Addressed	Learning Targets	
MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode and range. <i>Clarification 1: Numerical data is limited to positive rational numbers.</i> MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile and the maximum. Use this summary of the data to describe the spread and distribution of the data. <i>Clarification 1: Instruction includes describing range, interquartile range, halves and quarters of the data.</i> MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers and the range. <i>Clarification 1: Refer to K-12 Mathematics Glossary (Appendix C).</i>	Lesson 3	- I can find the mean, median, mode, and range of a given data set. - I can interpret the median, mode, and range of a given data set.
	Lesson 4	- I can determine the minimum, lower quartile, median, upper quartile, and maximum from a box plot. - I can determine the interquartile range from a box plot. - I can describe the spread and distribution of data from a box plot.
	Lesson 5	I can qualitatively describe and interpret a histogram within a real-world context.
	Lesson 6	- I can describe the shape of a histogram. - I can describe the spread of a data set given a histogram. - I can describe gaps, outliers, and clusters in a histogram.
	Lesson 7	- I can describe and interpret data represented on a line plot. - I can determine the similarities and differences between a line plot and a histogram.
	Lesson 8	- I can find and interpret mean and median of data sets. - I can use mean and median to compare data sets. - I can compare the distribution of data sets.

Section 3: Creating Data Displays (Lessons 9-10)

Benchmarks Addressed		Learning Targets
MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile and the maximum. Use this summary of the data to describe the spread and distribution of the data. <i>Clarification 1: Instruction includes describing range, interquartile range, halves and quarters of the data.</i> MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.	Lesson 9	- I can find key values given a box plot within a real-world context. - I can create a box plot to represent data.
<i>Clarification 1: Instruction includes collecting data and discussing ways to collect truthful data to construct graphical representations.</i> <i>Clarification 2: Within this benchmark, it is the expectation to use appropriate titles, labels, scales and units when constructing graphical representations.</i> <i>Clarification 3: Numerical data is limited to positive rational numbers.</i>	Lesson 10	I can create a histogram to represent data.



Vertical Alignment

Grade 6 Accelerated Unit 9
Data Sets



Grade 7 Accelerated Unit 13
Representing Data

Prior Course(s)	Course Level	Future Course(s)
MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range. MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.	MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data. MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range. MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range. MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.	MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

9.1 Types of Data

Lesson Introduction

 Benchmark	MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.		 Learning Target	I can classify data as categorical or numerical.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.		N/A	
 Essential Questions	- What is categorical data? What is numerical data? - What types of questions can you ask to collect categorical/numerical data? - What is a statistical question?			
 Lesson Narrative	Students will learn to classify statistical data as categorical or quantitative. Data will be presented numerically, in tables, or as graphs. Students will create statistical questions that generate numerical data. Student-gathered data will be used for examples.			
 Component Summary	9.1.1 Warm-Up (~5 min)	Students answer reflective questions related to how to use data <i>(SE p. 41)</i>	9.1.4 Guided Practice (5 min)	Classify whether data in scenarios is categorical or statistical and practice creating statistical questions <i>(SE p. 44)</i>
	9.1.2 Guided Instruction (15 min)	Review statistical questions, variability, and classifying data as categorical or numerical <i>(SE p. 42)</i>	9.1.5 Your Turn! (5 min)	Practice classifying variables as categorical or numerical <i>(SE p. 45)</i>
	9.1.3 Exploration (10 min)	Complete a class survey on a variety of questions and then review the class data set <i>(SE p. 44)</i>	9.1.6 Wrap-Up (~5 min)	Find the error made by a student classifying numerical and categorical data <i>(SE p. 46)</i>
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> - Data - Statistical question - Categorical data - Numerical data <u>Additional Terms:</u> N/A	(Optional) Highlighters (Optional) Chart Paper (Optional) Markers	N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may think that all data with numbers must be quantitative.	- Consider giving students some simple examples of surveys before beginning the Warm-Up. - When describing categorical data, include labels, names, colors, or examples. - Discuss how data is collected and what it can be used for. - For 9.1.3 Exploration question 2, share the data in a display for students.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 9.1.2 Guided Instruction provides an opportunity to engage students in a Notice and Wonder routine. Position students to notice the difference between the types of data and draw conclusions about how to classify data sets and variables.	MA.K12.MTR.1.1: Participate in Effortful Learning In this lesson, students are analyzing and classifying types of data. Encourage students to help and support each other through the lesson, specifically 9.1.3 Exploration and 9.1.4 Guided Practice. As students learn to identify statistical data, they will build perseverance thus enabling them to solve more challenging problems.
 Differentiate Instruction	This lesson provides auditory entry points through partner discussion and 9.1.2 Guided Instruction. Consider ways to provide visual entry points for learners to engage with the content, such as bar graphs. It may also be helpful for students to use different colored pencils or highlighters to label the types of data. This will provide a visual for students to make connections between data sets and the types of data.	
Lesson Notes:		

9.1.1 Warm-Up: Using Data

Component Overview: Students create survey questions and determine ways in which survey data can be used.



9.1.1 Warm-Up: Using Data

At the end of the school year, the school administration will ask students to participate in a survey about their favorite class.

1. What are two possible questions the survey could ask?

Answers will vary.

What is your favorite class?

How much time do you spend on homework?

2. Makayla thinks it would be helpful to ask students to rank their current classes on a scale of 1-6, with 1 being the students' favorite class and 6 representing their least favorite. Explain how this data could be useful.

Answers will vary.

The data could be useful because it could capture more information than just one class per student.

Students can compare their preferences with others.



Encourage a variety of question types. Students may adjust their suggestions for questions based on how they think the administration could use the data.

9.1.2 Guided Instruction: Types of Data

Component Overview: In this Guided Instruction, students review the formal vocabulary terms used to describe data sets. Students work in partners to create a statistical question and gather data. Students analyze the data to determine the level of variability and then review the different types of data. Students apply the definitions to classify types of data in a table.



9.1.2 Guided Instruction: Types of Data



A **statistical question** is a question that can be answered by collecting data. Often, there will be variability in the data.

Examples and non-examples of statistical questions are shown.

Statistical Questions	NOT Statistical Questions
- How many siblings do you have? - How tall are you? - What is your favorite song?	- What time does school start? - Who is the President of the United States?

- With your partner, discuss the differences between the two types of questions. What do you notice?

Statistical questions generate data that has many possible answers. Non-statistical questions have one correct answer.



Data are values that are collected together for reference or analysis.

- What do you think “variability in the data” means?

Answers will vary. Variability means there are variations, or differences, in responses.



Notice and Wonder

Engage the class in a Notice and Wonder routine to determine the differences between the two types of questions. Position students to consider what makes a question statistical (i.e., the questions can have many different responses, there is no right answer, etc.) and draw the conclusion that nonstatistical questions have one correct response.



For English Language Learners whose home language is Spanish, point out that “variability” in English translates to “variabilidad” in Spanish.

9.1.2 Guided Instruction: Types of Data (continued)

3. Work together to develop a statistical question. Use your question to gather five data values from your classmates. The answers to your question should all be numeric. Your classmates' responses are the data values.

a. Complete the table. *Answers will vary.*

Question				
<i>How much time do you typically spend on homework per night?</i>				
Data				
60 <i>minutes</i>	25 <i>minutes</i>	45 <i>minutes</i>	30 <i>minutes</i>	60 <i>minutes</i>

- b. Does the data from your question show a lot of variability or little variability?
Answers will vary based on question in Part A. Yes, there is a lot of variability.
- c. Explain how you determined your description of the variability.
The time spent on homework for the student with the longest time is over twice the time spent for the student with the shortest time.
4. Two statistical questions are shown.
Question 1: "How many siblings do you have?"
Question 2: "How many hours do you spend outdoors on a weekend?"
- a. Explain which question you would expect to have more variability.
Question 2
- b. Discuss your choice with your partner.
Answers will vary. It is reasonable to think that the students have about the same number of siblings, but some students may spend much more time than others outdoors on the weekend.



Students may struggle to understand what it means for a data set to be more variable. Ask clarifying questions:

- What might you look for to determine variability in a question?*
- Can you make a model or illustration to prove your thinking?*

9.1.2 Guided Instruction: Types of Data (continued)

Data can fall into two different classifications — categorical or numerical.



Categorical data is a type of data that is divided into groups. Categorical data are qualitative.

Numerical data, or quantitative data, is a type of data that is measurable. It is collected in the form of numbers.

5. In the table, classify the type of data described, or provide an example of data with that classification.

Data Description	Examples	Data Classification
m&m colors	red, blue, green, yellow, brown	<i>categorical</i>
number of pets a student has	0, 2, 1, 4, 8, 2, 2, 1	<i>quantitative</i>
zip code you live in	<i>Answers may vary.</i> 32940, 32605	<i>categorical</i>
<i>Answers will vary.</i> <i>favorite season</i>	<i>Answers will vary.</i> <i>fall, spring, winter, summer</i>	categorical
<i>Answers will vary.</i> <i>hours of sleep per night</i>	<i>Answers will vary.</i> 5, 6, 7, 8, 9	numerical



Consider having students create a Frayer Model with these vocabulary terms with examples and non-examples to solidify conceptual understanding. Students must understand the characteristics of each classification.



Students may confuse the classifications of data, especially if the categorical data can be quantified or represented numerically, such as with zip codes or phone numbers. For students already familiar with averages, point out that numerical data can be averaged, whereas categorical data with numbers such as zip codes does not have an average.

Ask clarifying questions to help students discern between the two:

- How can you describe numerical data? Categorical data?
- Can categorical data be represented numerically? Why?
- How can you measure numerical data?

9.1.3 Exploration: Class Survey

Component Overview: Students answer statistical questions. Once students have answered each question, collect and display the data for the class to see and review as a class. Then students categorize the questions by type of data.



9.1.3 Exploration: Class Survey

1. Answer the following questions. Write your answers in the table under the number that corresponds to each question.

- How many letters are in your first name?
- How old are you, in years?
- What is your favorite food?
- What is your favorite sport?
- How many languages do you speak fluently?
- What languages do you speak fluently?

Answers will vary.

Your Responses to Each Question					
a.	b.	c.	d.	e.	f.
6	11	pizza	soccer	2	English, Spanish

2. Review the display of the combined class data.

- Which data sets are categorical?
3, 4, 6
- Which data sets are numerical?
1, 2, and 5



Consider using technology to allow students to respond to the data in real time (example: Google survey, Formative, Kahoot, etc.). This will allow for a more easily available method of data collection and display.



Notice and Wonder

Position students to analyze the data through a Notice and Wonder routine.

- What do you notice about the data?
- Which questions, if changed, could alter whether the set is numerical or categorical?
- What do you wonder after looking at the data?

9.1.4 Guided Practice: Classifying Data

Component Overview: In this Guided Practice, students classify data as either categorical or numerical and explain how they know. Students then create their own statistical questions that will result in categorical data being collected and numerical data being collected.



9.1.4 Guided Practice: Classifying Data

1. A city parks department recently collected information about the breeds of dogs visiting a local dog park. They discovered that over 50 breeds of dogs visited the park on Saturday.

Determine if the data is categorical or numerical.
Explain your thinking.

The data is categorical because each answer represents a breed of dog, and the responses can be categorized as each breed.

2. At the beginning of the season, the football coach creates a roster with the players' names, ages, grade levels, and uniform numbers.

- a. Which of the variables are categorical?
players' names, uniform numbers, grade level

- b. Which of the variables are quantitative?
ages

- c. For each variable, explain how you determined the type of data.
Age generates numerical answers and variability of data. Players' names and uniform numbers are unique to each player. The responses for grade level can also be placed into categories so the data is categorical.



Encourage students to refer to previous components of the lesson or their Frayer Models if they are struggling to classify data.



Students may misclassify uniform number and grade level as numeric. Uniform number is a form of identification number, and grade level is a category.

9.1.4 Guided Practice: Classifying Data (continued)

3. What is the average weight of an elephant, in pounds?
 - a. What is your guess?
Answers will vary. 12,000 pounds.
 - b. Compare your answer with four other students.
What do you notice?
Answers will vary.
 - c. Is this categorical or numerical data?
numerical
4. Create a statistical question that will result in categorical data being collected.
Answers will vary. In what month were you born? What season do you like the most?
5. Create a statistical question that will result in quantitative data being collected.
Answers will vary. How tall are you? How many hours a week do you spend on social media?



As students create statistical questions for questions 4 and 5, consider walking around the room and choosing a few questions to share with the class. Write the questions on the board at the front of the class and prompt students to classify the resulting data as quantitative or categorical.

9.1.5 Your Turn!

Component Overview: Students work independently to classify variables and data sets as numerical or categorical.

9.1.5 Your Turn!

- Classify each variable as Categorical (C) or Numerical (N).

- Social media followers N
- Color of cars in the faculty parking lot C
- Shoe size N
- Cost of a drink N
- Zip code C
- Type of pets you have C



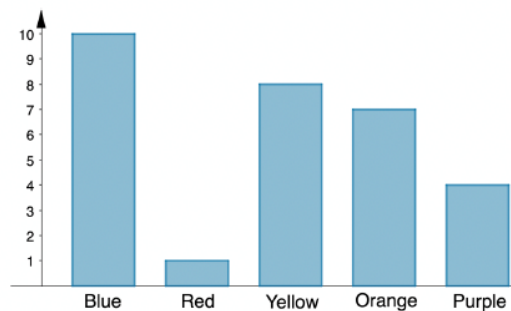
Students may struggle to explain their thinking in writing. Consider providing sentence stems to help students with their written response:

- "The responses will create categorical/numerical data because..."

- Will the question, "What is your favorite television or movie streaming service to use?" result in categorical or quantitative data? Explain how you made your choice.

The responses will create categorical data because the answers will be examples of streaming services, not numbers.

- Given this bar chart, classify the data as categorical or quantitative.



categorical

9.1.6 Wrap-Up: Error Analysis

Component Overview: Students complete an error analysis and correct a mistake in categorizing data.



9.1.6 Wrap-Up: Error Analysis

Bradley's teacher asked him to circle the numerical data in the table below.

Name	Student ID	Age	Hair Color	Grade Level
Rachel	1254866	10	Brown	6
Elijah	2548631	11	Brown	6
Chase	2645852	11	Blonde	6
Ben	2548654	10	Black	6

1. His teacher said that his response was incorrect. Correct his mistake and explain what mistake he made.
Student ID numbers are unique to each student, so they are categorical data.



For students who might struggle with an error analysis, consider scaffolding the question. Ask them first to solve the problem, explain their thinking, and then identify one to two errors a student may have made if they were to attempt a different method.

9.2 Statistical Questions

Lesson Introduction

 Benchmark	MA.6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.		 Learning Target	I can write statistical questions that generate numerical data.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.		N/A	
 Essential Questions	- What is a statistical question? - What is needed in a survey question to generate numerical data?			
 Lesson Narrative	In this lesson students will create statistical survey questions to gather data from the class. Students will classify the data as categorical or quantitative. Students may encounter non-statistical questions and have the chance to identify them and rewrite them as a statistical question.			
 Component Summary	9.2.1 Warm-Up (~5 min)	Create survey questions (<i>SE p. 47</i>)	9.2.3 Your Turn! (10 min)	Examples and non-examples of statistical questions (<i>SE p. 49</i>)
	9.2.2 Activity (25-30 min)	Gather data through survey questions (<i>SE p. 48</i>)	9.2.4 Wrap-Up (~5 min)	Reflect on ability to reach learning target (<i>SE p. 49</i>)
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Statistical question	(Optional) Chart paper (Optional) Chart makers (Optional) Computers	If using a form and/or spreadsheet application, consider teaching students how to use prior to the lesson.	

9.2.1 Warm-Up: Math Talk

Component Overview: In this Warm-Up, students will create questions to later survey the class with to gather statistical data. Consider reviewing student answers and having students revise their questions if they do not meet the criteria.



9.2.1 Warm-Up: Math Talk

In the spaces below, write three statistical questions that you could use today to collect data from your classmates. Be sure at least one of your questions will collect numerical data. *Answers will vary.*

1. How old are the students in my class?
2. How tall, in inches, are the students in my class?
3. How long does it take for my classmates to get to school in the morning?



Students may struggle to create statistical questions. Check student answers for appropriateness. Consider allowing students to brainstorm ideas for numerical data they can find out about their classmates before forming their questions.



MA.K12.MTR.7.1: Mathematics in Real-World Contexts
In this Warm-Up, students will create real-life questions to ask their classmates and gather data. In creating their own questions, students will feel more connected to the data.

9.2.2 Activity: Gallery Walk

Component Overview: In this Gallery Walk, students work with a partner to choose the best statistical question from the 9.2.1 Warm-Up to pose to the class. Each partner will choose a question and then post it in a location around the room for the rest of the class to answer. Ensure students understand where and how to post their questions. Then the class will rotate to answer each question. If you have a large class, consider placing students in groups and posting one question per group.



9.2.2 Activity: Gallery Walk

1. With your partner, discuss the following questions.

- a. How do you know the questions you wrote are statistical questions?
Questions are statistical if you can use them to collect data with variability.

- b. How can you determine which of the questions will generate numerical data?
The questions will yield a variety of numerical responses.

2. With your partner, determine the two best questions whose answers provide **numerical data** from the six questions you each created.

Determine which of these two questions each partner will use to collect data about your class.

Answers will vary.



Consider providing sentence stems for students to frame their thinking when discussing questions with their partner:

- “Questions are statistical if they...”
- “I know questions are statistical because they...”
- “Questions will generate numerical data if the responses are...”

9.2.2 Activity: Gallery Walk (continued)

3. Use your statistical question to collect some numerical data following the directions below. This data will be used in a later lesson, so keep it safe.
 - a. Write your statistical question in the top of the data chart that you chose in the previous problem.
 - b. Place the data chart in a location that is easily accessible by your classmates.
 - c. Follow your teacher's directions to answer each group's question.

Answers will vary.

My Statistical Question			
How tall in inches are the students in my class?			
Name	Answer	Name	Answer
1. Reid	64 inches	15.	
2. Kaylee	63 inches	16.	
3. Ava	60 inches	17.	
4. Sebastian	58 inches	18.	
5.		19.	
6.		20.	
7.		21.	
8.		22.	
9.		23.	
10.		24.	
11.		25.	
12.		26.	
13.		27.	
14.		28.	



Ensure students are clear on where to post their questions throughout the room and where and when to move to each question. Consider how you will signal the class to move on, how many students should be at each chart, and how each group should be moving around the room.

9.2.3 Your Turn!

Component Overview: In this *Your Turn!*, students give examples of statistical and non-statistical questions. Students explain the definition of a statistical question in their own words, then use the definition to justify whether sample questions are statistical or not.



9.2.3 Your Turn!

1. Give two examples and two counterexamples of a statistical question.

Examples of Statistical Questions	Counterexamples that are NOT Statistical Questions
<i>How often do you go to the public library?</i>	<i>Which state has the most public libraries?</i>
<i>What is the average age of U.S. presidents at their inauguration?</i>	<i>What is the age of the youngest U.S. president at the time of inauguration?</i>

2. Write the definition of a statistical question in your own words.

Answers will vary. Statistical questions can be answered by collecting data that varies.

3. The question, "How tall is the tallest man in the world, in inches?" is not statistical. Use your definition of a statistical question to justify why it is not a statistical question.

There is only one correct answer to the question. It does not elicit a variety of responses.

4. Decide if the question, "How many miles do you live from the school campus?" is statistical or non-statistical. Explain your decision.

It is statistical because you can use it to gather data with a variety of responses.



When discussing non-examples, encourage students to think of questions that could not have multiple answers. Students should also consider questions that are numerical but only have one response. Connect the word "statistical" with stats, like in sports.



Students may struggle to explain their responses in writing. To further scaffold, ask questions like:

- What makes a question a statistical question? Why?
- How many answers does the question about the man's height have? Will it have more than one?
- Does everyone live the same distance from school? What does that tell you about the question?

9.2.4 Wrap-Up: Reflection

Component Overview: In this Wrap-Up, students will discuss how they felt throughout the lesson trying to reach the learning target. Consider using this as a guide for the next day to see what to focus on or reteach with each student.



9.2.4 Wrap-Up: Reflection

1. What is the most difficult part about writing statistical questions?

Answers will vary. Figuring out if the question elicits a numerical response or not.

2. How can it be determined whether or not a statistical question can be answered with numerical data?

*Answers may vary.
A statistical question can be answered with numerical data if data has numbers that are not used as categories.*

3. What was the most surprising response to your statistical question? Why?

Answers will vary. Responses should apply to the statistical question the student created.



Some students may struggle to explain their reflections in writing. Consider alternate means of assessment. Students could partner up and answer the questions verbally as you circulate and listen for common themes.

9.3 Finding and Interpreting the Mean, Median, Mode, and Range

Lesson Introduction

 Benchmark	MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range.		 Learning Targets	- I can find the mean, median, mode, and range of a given data set. - I can interpret the mean, median, mode, and range of a given data set.	
 Benchmark Coherence	Building On			Working Toward	
	MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.			MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.	
 Essential Questions	- How can you find and interpret the mean, median, mode, and range of a given data set? - What does the mean/median/mode/range of a data set tell you? - How does context impact the interpretation of mean/median/mode/range of a data set?				
 Lesson Narrative	In this lesson, students will explore and find the mean, median, mode, and range of a given data set. Students will interpret their findings to describe the data sets. Students will work with partners to explore the concepts of mean, median, and mode.				
 Component Summary	9.3.1 Warm-Up (~5 min)	Compare lengths of lines (<i>SE p. 50</i>)		9.3.4 Guided Instruction (10-15 min)	Find mean, median, and mode using definitions (<i>SE p. 52</i>)
	9.3.2 Exploration (5-10 min)	Discover how to redistribute groups of a square evenly (<i>SE p. 51</i>)		9.3.5 Try It (5 min)	Practice finding mean and median (<i>SE p. 54</i>)
	9.3.3 Collaboration (5-10 min)	Find the average (mean) of a given set of numbers (<i>SE p. 51</i>)		9.3.6 Exploration (5-10 min)	Practice finding mode and range (<i>SE p. 54</i>)
	9.3.7 Wrap-Up (~5 min)	Reflect on understanding of mean, median, mode, and range (<i>SE p. 56</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Mean - Median - Mode - Range - Measures of center - Measures of variability <u>Additional Terms:</u> N/A		(Optional) Calculator (Optional) Computer		Consider predetermining partners before the lesson to help with matching students appropriately.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse mean and median or mode and median. - Students may think the mean or median must be one of the values in the data set. - Students may try to find the median before ordering the data set. - Students may forget to divide by the number of data values after finding their sum to determine the mean.	- Students may feel overwhelmed with finding four different measurements. Consider providing graphic organizers or chunking information. - Encourage students to write out all of the numbers in a data set in order from least to greatest every time they deal with the set to make sure they do not miss a number - Consider providing students with extra paper to solve and give more room to think. - Encourage students to show their work on every problem to avoid basic mistakes. - If time is an issue, then consider assigning 9.3.5 Try It for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share In 9.3.2 Exploration, students complete questions 1 and 2 independently. Then students discuss their responses and strategies with a partner. These discussions help students gain confidence and understanding.	MA.K12.MTR.7.1: Real-World Contexts Data throughout this lesson is presented through real-world contexts. Encourage students to connect the content of the lesson to their everyday lives by considering when summarizing data using the mean, median, mode, or range might be helpful. This will build investment in the mathematics.
 Differentiate Instruction	Notice the opportunity to use manipulatives and visuals during 9.3.2 Exploration. Visual entry points are available through the display of data in graphs and charts. Encouraging the use of manipulatives in this section will provide an additional kinesthetic entry point for learners to engage with the content. This is particularly helpful as students use manipulatives to create equal groups and compare this to the algorithm they may have used to find the mean. This connection deepens understanding of how to find the mean and can help students understand the context, particularly when determining the number needed to distribute evenly amongst groups. Students will see how certain data, such as animals or humans, cannot be split into fraction or decimals.	
Lesson Notes:		

9.3.1 Warm-Up: Stripes

Component Overview: In this Warm-Up, students will compare stripes on four groups of cubes. Students will discuss the similarities and differences between the sizes.



9.3.1 Warm-Up: Stripes

A picture of four sets of 16 cubes is shown.



For those who struggle with spatial reasoning, consider allowing students to cut out each colored strip to compare them more easily. Consider having students imagine how triangles could make up each line to also get a better visual on how long each is.

1. Which set of cubes has a set of stripes that covers the most area?

The set in the bottom left corner with the green stripes is the set with stripes that cover the most area.

2. Explain how you determined your answer.

Answers may vary. It has the most stripes, so they logically cover the most area.

3. Which set of cubes has a set of stripes that covers the least area?

The set in the top right corner with the blue stripes is the set with stripes that cover the least area.

4. Explain how you determined your answer.

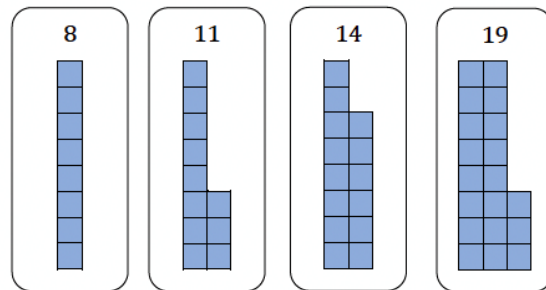
Answers may vary. If you put the blue stripes together, they will still be shorter than the one red stripe.

9.3.2 Exploration: Sharing Squares

Component Overview: In this Exploration, students evenly distribute squares into groupings. Students translate this into working with adding and dividing numbers to find the mean.



9.3.2 Exploration: Sharing Squares

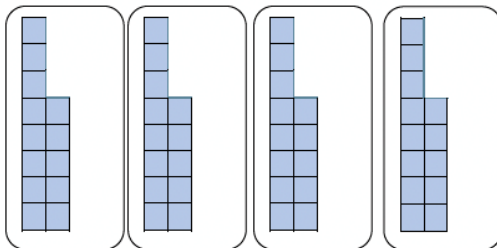


Consider providing square manipulatives for students to help understand the best way to distribute the squares evenly into four parts. This will provide a kinesthetic entry point.

1. Estimate the number of squares that will be in each group if they were distributed evenly between the four groups.

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2. Use the diagram to redistribute the squares so that each of the four groups has the same number of squares.



3. Compare diagrams with your partner. Discuss how you each determined the number of squares that belong in each group. Summarize your discussion.

Answers will vary. Add all the blocks together and then divide by the number of cards, 4. I moved the blocks around from the columns with 14 and 19 so that the number in each became more even.



Students may struggle to explain how they determine the number of squares that belong in each group. To provide further guidance for discussion, ask students to consider the following questions:

- What did you do first to consider how to divide the squares?
- What does it mean to distribute evenly? Will each group have the same number? Why?
- How does the number of groups impact your answer?

9.3.3 Collaboration: Mean of Data

Component Overview: In this Collaboration, students will work together to find the mean of a data set using the given information. This page requires a heavy amount of reading, which may need to be broken down for struggling readers and ELLs. This section can be used as a guide for notetaking.



9.3.3 Collaboration: Mean of Data

By redistributing the squares in the diagram, the **mean** of the data set was found.



The **mean** is the arithmetic average of a set of numbers. It is a measure of central tendency.

Data in a numerical set is summarized using different measures that are interpreted to describe the data. One such measure used to describe quantitative data sets is the mean.

To calculate the mean, add all the values in the data set and then divide by the total number of data values.

1. A marine biologist is studying turtles and wants to calculate the average length of time a Red-Eared Slider, a popular breed of turtle, can remain underwater. The time, in minutes, that ten Red-Eared Sliders remain underwater are 75, 300, 225, 400, 250, 195, 320, 45, 165, 410.
 - a. To calculate the mean time spent underwater, start by finding the sum of the data values.
The sum is 2,385.
 - b. Then divide the sum by 10, because there are ten turtles.
 - c. In this sample, Red-Eared Slider turtles remain underwater for an average of 238.5 minutes.



Measures of center are numerical values used to describe the overall clustering of data in a set, or the overall central value of a set of data. The three most common measures of central tendency are the mean, median, and mode.



Turn and Talk

This section is intentionally scaffolded to build conceptual understanding of what mean is and why it is used. Have students engage in a Turn and Talk routine to restate the definition in their own words. This will ensure understanding over rote memorization.



Consider providing a graphic organizer to frame student thinking and help set up the proper way to find the mean of a set of numbers. Students may struggle with seeing numbers vertically and may need to view them horizontally.

9.3.4 Guided Instruction: Mean, Median, and Mode

Component Overview: In this Guided Instruction, students begin by filling in the definitions using terms from a word bank. Go through the definitions with the class to ensure students understand what each term means before moving on. Students then find the mean, median, and mode of a data set and describe the steps to find each of them.



9.3.4 Guided Instruction: Mean, Median and Mode

- For each definition, fill in the blank with one of the words from the bank.

median	mode
mean	range



The mean is the arithmetic average of a set of numbers. It is a measure of central tendency.



The range is the difference between the highest data value and the lowest data value in a data set. It is a measure of variation.



The mode is the value found most often in a set of numbers. There may be no, one, or more than one mode in the data set. It is a measure of central tendency.



The median is the middle of an ordered list of the values. If the list has an odd number of values, it is the middle value of that list. If the list has an even number of values, it is the average of the two middle values. It is a measure of central tendency.



Fill in the blanks for the glossary terms with the class. Begin by reading a definition to the class. Then have students talk with a partner to find and define the term in their own words. Review the correct answers as a class.



Allowing students to discuss each glossary term with a partner provides an auditory entry point to engage learners. Consider providing visual entry points, such as a histogram and data set with each term labeled, to engage visual learners.

9.3.4 Guided Instruction: Mean, Median, and Mode (continued)

The number of pets ten students in Tyler's class have is reflected in the table.

Number of Pets									
1	0	2	1	10	0	2	0	8	12

- Describe steps to find the median.
*Order the numbers from least to greatest.
Find the middle number. If there is an even number of entries in the data set, find the average of the two middle numbers.*
- The median number of pets for the ten students is 1.5.
- The mode of the data set is 0.
- Write an explanation of what the mode of the data set means for the context.
Answers will vary. The most common response is zero pets.
- Describe the steps needed to find the average number of pets owned by the ten students.
Find the sum of all the values in the data set. Then divide the sum by the number of values in the data set, 10.
- The mean number of pets for the ten students is 3.6.
- Complete the statement about the mean of the data set.
"Each student in Tyler's class has about 3 or 4 pets."



Students may struggle to describe the steps to find the median. Provide support through questioning:

- What is the first thing you need to do to the data set to begin to find the median? Why?
- After the numbers are in order, how can you find the median? Is the number of data points significant? Why?

9.3.5 Try It: Mean, Median, and Mode

Component Overview: Students work independently or in pairs to interpret data in a table. Students find the mean and median of the data set and describe what they mean in the context of the question. Circulate as students work to complete this section and make note of any misconceptions to address either with individual students or as a class.



9.3.5 Try It: Mean, Median, and Mode

1. Nine adults boarding a cruise ship at Port Canaveral in Florida were randomly surveyed on the total number of vacations they have taken in the last five years. Their totals are summarized below.

Number of Vacations Taken								
5	4	3	9	1	4	0	6	3

- a. What does the value of the median represent in the context?
Half of the vacationers have taken more than the median number of vacations in the last five years and half have taken fewer.
- b. What is the median number of vacations taken by the adults surveyed?
4
- c. Determine the mean of the data. Then, interpret the mean in terms of the context by writing a sentence to summarize. Round to the hundredths.
The mean is 3.89 vacations. The people surveyed have taken an average of 3.89 vacations in the last five years.



As an extension, consider allowing students to discuss how the answers might change if they asked people on the street or a more general location instead of those boarding a cruise ship.

9.3.6 Exploration: Understanding Mode and Range

Component Overview: In this Exploration, students will work with a partner to discover the purpose of the mode and range using the definitions from the 9.3.4 Guided Instruction.



9.3.6 Exploration: Understanding Mode and Range

Work with your partner to complete the following.

1. The ☐ mean ☐ median ☒ mode ☐ range is the value occurring most often in

a set of data, which can be used to describe trends in data.

2. While shopping over the last month, Shep collected the price for a jar of pickles from eight stores he visited.

Price for a Jar of Pickles			
Store A	\$1.66	Store E	\$1.99
Store B	\$2.99	Store F	\$1.99
Store C	\$1.49	Store G	\$1.50
Store D	\$1.99	Store H	\$1.99

- a. What does a jar of pickles cost at most stores?
\$1.99
- b. This value is the ☐ median of the data set. ☒ mode of the data set.
- c. What is the most Shep will spend on a jar of pickles?
\$2.99
- d. What is the least Shep will spend on a jar of pickles?
\$1.49
- e. The range of a data set describes how spread out the data is. Determine the range in price for a jar of pickles.
 $\$2.99 - \$1.49 = \$1.50$



To provide an additional visual entry point, consider encouraging students to represent the data on a number line for questions 2 and 3. This will clearly display the spread of the data and allow students to visually see the range and mode.

9.3.6 Exploration: Understanding Mode and Range (continued)

3. Shep also began collecting the price of his favorite cereal to determine where he would get the best price.

Price for a Box of Cereal			
Store A	\$3.79	Store E	\$3.99
Store B	\$3.89	Store F	\$3.89
Store C	\$3.99	Store G	\$4.29
Store D	\$3.89	Store H	\$4.59

- The mode price of his favorite box of cereal is **\$3.89**.
 - What is the most Shep will spend on a box of cereal?
\$4.59
 - What is the least Shep will spend on a box of cereal?
\$3.79
 - The range for the price of a box of cereal is **\$0.80**.
 - The range describes the

☐ center
☒ variability

 of the data set.
4. With your partner, discuss whether the price for pickles or the price for cereal has more variability in pricing. Summarize the discussion.

The price for pickles has more variability as seen by pickles having the larger range of prices.



Measures of variability are numerical values that measure how much a data set varies from a central value.



The range is a **measure of variability**. By comparing the ranges for two or more data sets, the data set that has more variability can be determined by finding which has a larger **range**.



Think-Pair-Share

Implement a Think-Pair-Share routine before students complete question 4. This will give students a chance to consider what it means to have more variability and formalize their responses with a partner before writing. Circulate as students discuss and bring the class together to address any misconceptions that may arise. Students may not connect that a larger range indicates more variability. This is an important misconception to address with the class.

9.3.7 Wrap-Up: Reflection

Component Overview: In this Wrap-Up, students will reflect on their understanding of the lesson, what areas they excel in, and what areas they need more support and help with.



9.3.7 Wrap-Up: Reflection

1. Read each of the “I can” statements below. Check the boxes that best explain your understanding.

Answers will vary.

- ☐ I can find and interpret the median, mode, and range of a given data set.
- ☐ I understand how to find the mean, median, and mode of a data set.
- ☐ I need support on how to find the mean, median, and mode of a data set.
- ☐ I understand how to find the range of a data set.
- ☐ I need support on how to find the range of a data set.
- ☐ I understand how to interpret measures of central tendency.
- ☐ I need support on how to interpret measures of central tendency.

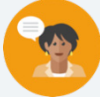


Consider using Google Forms to survey students to identify who needs the lesson to be retaught or reinforced.

9.4 Interpreting Box Plots

Lesson Introduction

 Benchmark	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data.		 Learning Targets	- I can determine the minimum, lower quartile, median, upper quartile, and maximum from a box plot. - I can determine the interquartile range from a box plot. - I can describe the spread and distribution of data from a box plot.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.	
 Essential Questions	- How do you interpret data from a box plot? - How does a box plot show range, minimum, maximum, median, and quartiles? - How would you find interquartile range from a box plot?				
 Lesson Narrative	This lesson explores how to use a box plot to interpret a set of data. Students construct and use a box plot to answer key questions about data. Students find the range, minimum, maximum, median, and quartiles of a data set as well as use the information to find the interquartile range.				
 Component Summary	9.4.1 Warm-Up (~5 min.)	Examining and discussing similar numbers (<i>SE p. 57</i>)	9.4.4 Guided Practice (5 min.)	Determining interquartile range (<i>SE p. 60</i>)	
	9.4.2 Exploration (10 min.)	Exploring a box plot (<i>SE p. 57</i>)	9.4.5 Try It (5-10 min.)	Using box plots to understand data (<i>SE p. 61</i>)	
	9.4.3 Guided Instruction (10 min.)	Learning about box plots and interquartile range (<i>SE p. 59</i>)	9.4.6 Your Turn! (5 min.)	Practicing working with box plots (<i>SE p. 62</i>)	
	9.4.7 Wrap-Up (~5 min.)	Describing box plots and how to use them (<i>SE p. 63</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Box plot - Quartile - Interquartile range <u>Additional Terms:</u> N/A		(Optional) Chart paper (Optional) Set of data for students to use to create a box plot		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse the median of the data with its mean. - Students may struggle to understand the purposes, placements, and meaning of the quartiles. - Students may confuse the range of a box plot with its interquartile range and how to find each from the data display.	- Highlight how to find the median of the data from a box plot. Prompt students to notice that the mean is not reflected in a box plot. - On chart paper, create a list of how to interpret each part of a box plot and what it means. Display this list for students to refer to throughout the lesson. - Highlight the different values used to calculate range and interquartile range. Prompt students to discuss how and when it would be helpful to use each one in a real-life context.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 9.4.2 Exploration positions students to notice and interpret the details of box plots. Consider applying Notice and Wonder routines for each box plot, positioning students to connect the data values and their representation on the box plot.	MA.K12.MTR.3.1: Mathematical Fluency Students will be working to create and interpret box plots. Clarifications within this standard emphasize accurately completing mathematical tasks and adapting procedures to new contexts. Provide students with the opportunity to practice the processes for interpreting a box plot and to reflect on the processes they use.
 Differentiate Instruction	Box plots offer visual entry points for students as they interpret and analyze data sets. Reference the effectiveness of multiple representations (MA.K12.MTR.2.1) when interacting with the data throughout the lesson. Using colors to label the different sections of the box plots provides an additional visual for students to use to interpret the data and connect the visual representation of the box plot with the data set.	
Lesson Notes:		

9.4.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students examine four numbers—a percentage, a fraction less than 1, a fraction greater than 1, and a repeating decimal—to determine which one is not like the others. Students will then explain why that number differs from the others. Students can accurately answer that any of the numbers does not belong as long as they can support their answer with solid, mathematical reasoning.



9.4.1 Warm-Up: Which One Doesn't Belong?

Examine the four numbers.

33%	$\frac{1}{3}$
$\frac{5}{3}$	$0.\overline{6}$

- Which of these numbers does not belong? Explain your thinking.

Answers will vary.



Challenge students to create additional groups of four numbers in which each number could logically not belong. Have students switch their numbers with another student or small group to discuss.

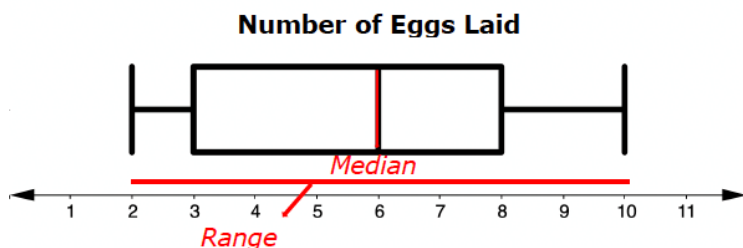
9.4.2 Exploration: Exploring Box Plots

Component Overview: Students will examine a box plot and use it to answer questions about the data expressed in it. Students may work with a partner on the entire activity or in small groups.



9.4.2 Exploration: Exploring Box Plots

1. A farmer recorded the number of eggs her hens laid in one week. The data set was the total number of eggs each of her 53 hens laid during the week. Below is a box plot representing the number of eggs laid by each hen during the week.



With your partner use the box plot to complete the statements.

- a. The smallest number of eggs laid by the hens is 2.
- b. The largest number of eggs laid by the hens is 10.
- c. Find another pair of partners and compare the smallest and largest number of eggs laid. Discuss how you determined the values from the box plot. If your values are different, come to a consensus on the smallest and largest number of eggs laid by the hens, and circle these values on the box plot.

Answers will vary. The smallest value is the first number in the 5-number summary, and in this scenario the smallest value is 2 and the largest value is 10.

Smallest Number of Eggs - 2

Largest Number of Eggs - 10



Provide scaffolded questions to bridge instructional gaps as students explore in groups:

- What is the lowest number in the data set? What does this represent? How is this value shown in the box plot?
- What is the greatest number in the data set? What does this represent? How is this value shown in the box plot?

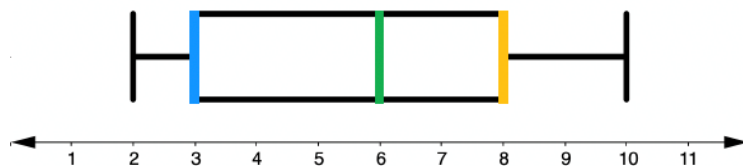


Students may struggle to interpret the box plot, but that is the purpose of the exploration – to provide students an opportunity to explore an unfamiliar data display and use what they know to begin interpreting its meaning. Consider ways to encourage and support your students as they work through the process with their partners or small groups. Consider also the purposeful grouping used for this activity to ensure rich discussions can be added in each group.

9.4.2 Exploration: Exploring Box Plots (continued)

- d. Label the range for the number of eggs laid on the box plot.
Range = Maximum - Minimum = $10 - 2 = 8$
- e. With your partner, conjecture the median number of eggs laid. Label this number as "median" on the box plot.
Labeled on box plot. The median is the middle value of a data set. It is the line in the middle of the box, 6.
- f. The first quartile is highlighted in blue, the second quartile is highlighted in green, and the third quartile is highlighted in yellow.

Number of Eggs Laid



If the second quartile is the median, M , write a conjecture with your partner about how the first and third quartiles are each determined.

Answers will vary. The first quartile is the first quarter of data, so it is the middle of the first half of data. The third quartile is the third quarter of data, so it is the middle of the upper half of the data values when ordered from smallest to largest.

2. Work with your partner to complete the statement.
A **box plot** divides data into 4 equal parts that each contain roughly 25% of data from a given data set.
3. Explain why you chose the value you used to complete the statement.
Answers will vary.



Notice and Wonder

Engage students in a Notice and Wonder routine to begin to analyze and draw conclusions about the value of representing data in box plots. Encourage students to notice similarities between their previous use of range and median and how they are used in box plots. Prompt students to think about why median, not mean, is part of a box plot.

9.4.3 Guided Instruction: Interpreting Box Plots

Component Overview: Students label parts of a box plot and formalize how to interpret it following their exploration. Students use the information in the box plot to determine the interquartile range. The Guided Instruction provides entry points for both visual and auditory learners. This is an opportunity to formalize what students explored in the previous component to ensure understanding of how to interpret data represented in a box plot.



9.4.3 Guided Instruction: Interpreting Box Plots

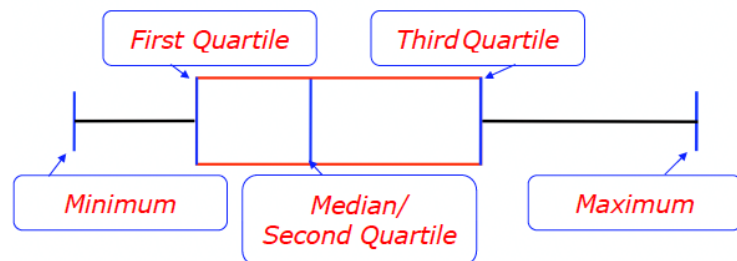


A **box plot** is a plot displaying the spread or distribution of a data set using a five-number summary: the minimum, lower quartile (first quartile), median, upper quartile (third quartile), and maximum. It is also called a box-and-whisker plot.



Consider adding numbers to the box plot in the 9.4.3 Guided Instruction if you think it will make the activity less abstract for students, or refer to the box plot they explored in the 9.4.2 Exploration. Use the data points to help students determine the minimum, maximum, median, quartiles, and interquartile range.

1. Label the maximum, minimum, median, first quartile, and third quartile on the box plot.



2. Complete the statement.

The **first quartile (Q_1)** is the middle value of the data

that are ☐ less than the median.

- ☐ greater than
- ☐ equal to

The **third quartile (Q_3)** is the middle value of the data

that are ☐ less than the median.

- ☐ greater than
- ☐ equal to

9.4.3 Guided Instruction: Interpreting Box Plots (continued)



For a data set with median, M , the first **quartile** is the median of the data values less than M , and the third **quartile** is the median of the data values greater than M . The second **quartile** is the median, M .

3. Complete the statement.

The measure of central tendency used to create a box plot is the _____, which divides the data set in half.

- ☐ mean,
- ☒ **median,**
- ☐ mode,

A number line often accompanies a box plot and is used to determine the values associated with each part of the plot.

Another measure of variability like **range**, the **interquartile range (IQR)** can be found by determining the difference in the third quartile from the first quartile.



A measure of variation in a set of numerical data, the **interquartile range** is the distance between the first and third quartiles of the data set.



$$\text{IQR} = (Q_3) - (Q_1)$$

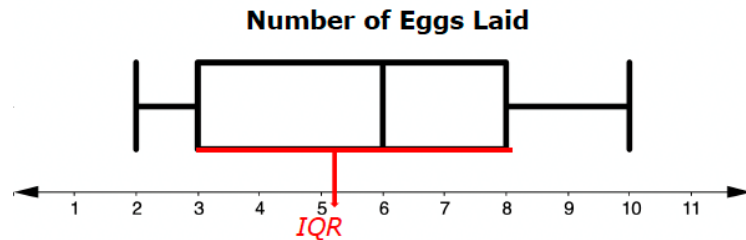


Encourage students to examine range and interquartile range by asking the following questions:

- How do you find the range? What does it represent?
- How do you find the interquartile range? What does it represent?
- How do range and interquartile range differ? How are they similar?

9.4.3 Guided Instruction: Interpreting Box Plots (continued)

4. Revisit the box plot of the eggs laid by a farmer's hens.



- a. What is the interquartile range (IQR) for the box plot of the context?

$$\text{Interquartile Range (IQR)} = Q_3 - Q_1$$

$$\text{Interquartile Range (IQR)} = 8 - 3$$

$$\text{Interquartile Range (IQR)} = 5$$

- b. Label the IQR on the box plot.

Shown on box plot.



Encourage students to think about the term “interquartile” and its relationship to the prefix “inter-,” meaning “between or among.” Discuss that the interquartile range is the range between the range created by the minimum and maximum values and also the range between the first and third quartiles.



Consider using different colors to highlight the different sections of the box plot. Use the same colors to show the calculations for how to arrive at the data points.

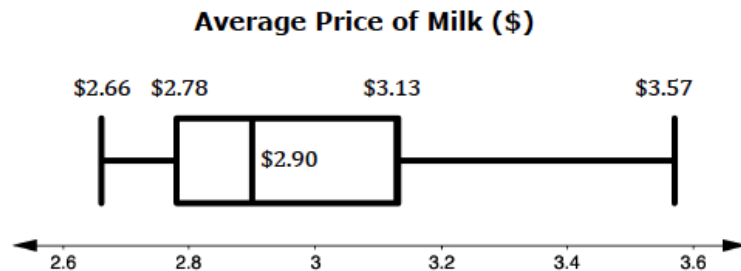
9.4.4 Guided Practice: Interpreting Box Plots

Component Overview: Students determine the range and interquartile range of the data represented by the box plot. Through the box plot, students can examine a real-world example of how to display data.



9.4.4 Guided Practice: Interpreting Box Plots

1. The average price of milk, in dollars (\$), for each month from April 2000 to November 2006 was used to create the box plot shown.



Use this section to focus students on range and interquartile range. If needed, review how to find the minimum, maximum, and quartiles as a vehicle to determining the values of range and interquartile range.

- a. What is the range of the data?
 $3.57 - 2.66 = 0.91$
- b. Determine the interquartile range.
 $3.13 - 2.78 = 0.35$
- c. Does the average cost of milk from April 2000 to November 2006 vary a lot or a little? Explain your reasoning.
Answers will vary. The range of nearly one dollar is almost half of the cost of milk in 2000, so that may be seen as a large variation.

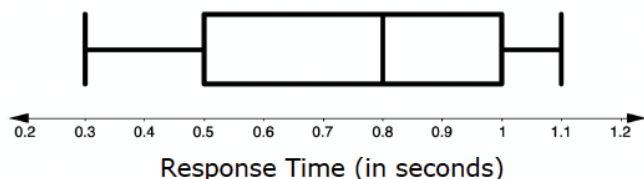
9.4.5 Try It: Interpreting Box Plots

Component Overview: Students can synthesize their findings in this activity. In the Try It, students will be utilizing all the knowledge and skills they have previously learned about box plots.



9.4.5 Try It: Interpreting Box Plots

- The box plot displays data on the response times of 100 mice when seeing a flash of light.



Complete the statements.

- The fastest response time is 0.3 seconds.
 - The slowest response time is 1.1 seconds.
 - The middle response time is 0.8 seconds.
 - The lower quartile is 0.5 seconds.
 - The upper quartile is 1 seconds.
 - The range is 0.8 seconds.
 - The interquartile range is 0.5 seconds.
- With your partner, use the values to write a description of the spread and distribution of response times.
Answers will vary. The mouse that responded the slowest to the flash of light was 1.1 seconds. The fastest response to the flash of light was 0.3 seconds. The range of difference between the fastest and the slowest time was 0.8 seconds. This is a large variation since the range is much larger than the fastest time.



"Fastest" is sometimes confused with "the largest," so students may incorrectly label the fastest as the maximum instead of the minimum time.



Provide scaffolded questions to encourage students to further examine the box plot:

- What is the range of the data? How would you find it?
- What is the first quartile? How would you find it?
- What is the third quartile? How would you find it?
- How do you use those values to find the interquartile range?

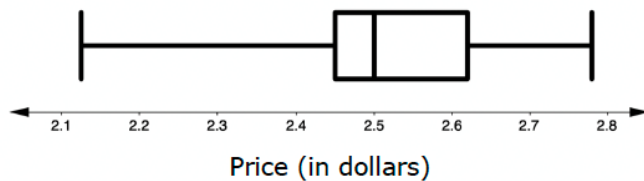


Consider allowing students to work with partners as needed. Partners can work together on each item or divide the work in half and then explain their findings to each other.

9.4.5 Try It: Interpreting Box Plots (continued)

3. The U.S. Energy Information Administration reports weekly average gas prices for the United States. The box plot represents the weekly average gas prices in 2019.

2019 Weekly Gas Prices



What percentage of weeks had an average gas price of at least \$2.50?

50%



Encourage students to more closely examine the box plot:

- Which value represents the highest gas price? How is this represented on the box plot?
- Which value represents the lowest gas price? How is this represented on the box plot?
- What is another name for the middle response time?

9.4.6 Your Turn!

Component Overview: Students continue working with box plots. The additional practice provides students with the opportunity to continue the processes for interpreting a box plot.



9.4.6 Your Turn!

1. Mrs. Henry's 3rd period English class took a spelling test with 12 words on it. A box plot was created from the number of questions students got correct on the test.

Number of Correct Questions



Complete the statements.

- a. The lowest score is 1.
- b. The highest score is 12.
- c. The middle score is 7.
- d. The lower quartile is 2.
- e. The upper quartile is 9.
- f. The range is 11.
- g. The interquartile range is 7.
- h. Write a description of the spread and distribution of the scores.

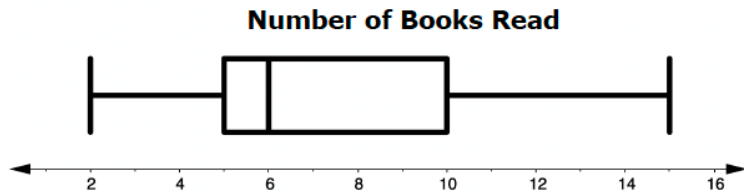
Answers will vary. The student with the lowest number of correct questions in Mrs. Henry's class answered only 1 question correctly and the student with the most correct answers answered all 12 correctly. This is a range of 10 questions. The middle 50% of students answered between 2 and 9 questions correctly, which is the interquartile range of 7 questions. In either case, this shows that the spread of distribution is large, there is large variation in the number of questions answered correctly.



Challenge students to describe the spread and distribution of scores in as much detail as possible. Ask students to think about why the data is distributed as it is and how this may vary in different situations.

9.4.6 Your Turn! (continued)

2. The librarian at Franklin Middle School surveyed students when they returned from summer break on how many books they had read. She created a box plot to represent the results of her survey.



What percentage of students read between:

- 5 and 10 books over the summer? 50%
- 10 and 15 books over the summer? 25%
- 5 and 6 books over the summer? 25%
- 2 and 10 books over the summer? 75%



Consider an extension project of your own classroom or school in which students in groups must collect the data, construct a model, and interpret the data and their findings as they continue working through this unit.

9.4.7 Wrap-Up: Cool Down

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' ability to construct a box plot and describe its components.



9.4.7 Wrap-Up: Cool Down

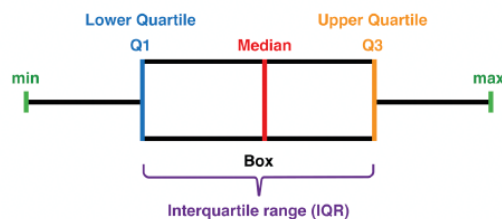
Jamal missed class today and he needs help with what he missed.

- Write a note to Jamal letting him know what a box plot is and what it is used for. Draw a picture or diagram that may help him better understand.

Answers will vary.

Dear Jamal,

You missed a fun class today on box plots. Box plots display the 5-number summary – minimum, first quartile, median/second quartile, third quartile, and maximum. The box plot has four parts that each represent 25% of the data, and it does not matter how big the part is, it still represents 25% of the data values. A box plot can be used to observe the distribution of the data. Here is a picture of a box plot to help:



For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.

9.5 Describing and Interpreting Histograms – Part 1

Lesson Introduction

 Benchmark	MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.		 Learning Target	I can qualitatively describe and interpret a histogram within a real-world context.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.	
 Essential Questions	- What is the purpose of a histogram? - How do you read a histogram? - How can you use a histogram to analyze data?			
 Lesson Narrative	This lesson is Part 1 of 2 in which students interpret histograms. Students begin by formally defining what a histogram is and then use a histogram to answer questions about a data set. Students continue to interpret data displayed in histograms and analyze the spread of the data to determine typical values and where measures of center can be found on the display.			
 Component Summary	9.5.1 Warm-Up (~5 mins)	Examine why statistics are important (<i>SE p. 64</i>)	9.5.4 Try It (5-10 mins)	Analyze a histogram (<i>SE p. 67</i>)
	9.5.2 Guided Instruction (5-10 mins)	Use a histogram to understand data (<i>SE p. 64</i>)	9.5.5 Your Turn! (5 mins)	Interpret a histogram (<i>SE p. 68</i>)
	9.5.3 Exploration (10 mins)	Interpret the relationship between a data table and a histogram (<i>SE p. 65</i>)	9.5.6 Wrap-Up (~5 mins)	Self-assessment of learning target (<i>SE p. 68</i>)
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> Histogram <u>Additional Terms:</u> N/A	(Optional) Colored pencils	9.5.1 Warm-Up contains a video to accompany the question prompt.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse histograms and bar graphs. - Students may struggle to understand the concept of frequency.	- Consider using the same set of data to create a histogram and a bar graph. Encourage students to notice the differences and similarities in the graphical representations. - Have students pick a color: red, blue, yellow, or green. Determine how frequently each color is chosen as a way to demonstrate frequency to students.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 9.5.2 Guided Instruction invites students to use a histogram to analyze a set of data. Consider applying Notice and Wonder routines to the histogram, encouraging students to discover the relationship between the data set and the representation of it in the histogram.	MA.K12.MTR.7.1: Real-World Contexts Students will be working to represent and analyze real-world data in a histogram. Clarifications within this standard emphasize using models and methods to represent everyday experiences. Utilize this lesson to provide opportunities for students to use histograms to investigate data.
 Differentiate Instruction	This lesson focuses heavily on the visual representation of data through histograms. Representing data sets graphically allows students to more easily make connections between statistical measures and the data set. Encourage students to color-code different bars in the histogram and draw lines to the corresponding frequency to provide additional visual and kinesthetic entry points. Partner discussion and reading questions aloud provide auditory entry points for students to engage with the content as well. Consider points in the lesson when students who may be struggling could benefit from the added support of partner- or teacher-led discussion.	
Lesson Notes:		

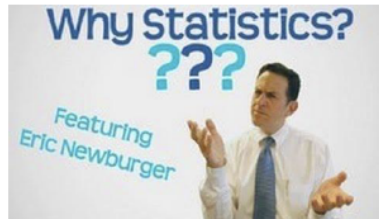
9.5.1 Warm-Up: Why Statistics?

Component Overview: Have students watch the video before reading the description and answering the prompt. Students will then consider additional reasons why statistics are important and how they can be utilized in daily life.



9.5.1 Warm-Up: Why Statistics?

Eric Newburger relates the need for studying statistics to the need for using a microscope. He discusses how a microscope is used to study small things. It's also needed to study things like food poisoning at a larger level to aid in understanding diseases and how viruses spread.



In Mr. Newburger's work, he explains you can take a big thing and shrink it down small enough to study and to understand. That's the study of statistics. And when you want to understand something that is too small, you can use a tool (like the microscope), which is "why statistics."

- List two reasons why statistics are important.
Answers will vary. There is a need for statistics because it allows us to present data and analyze data. Statistics also allows us to make predictions based on trends found in the data.



Do you know someone who is a statistician or works in data management? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around statistics as a career.

9.5.2 Guided Instruction: Histograms

Component Overview: Begin this section by reviewing the definition of a histogram as a class. Then have students work in pairs to analyze and interpret the data presented in the histogram about shark lengths.



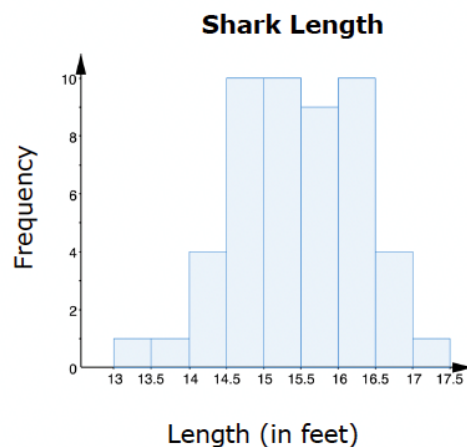
9.5.2 Guided Instruction: Histograms

Numerical data can be displayed graphically in a variety of ways. One type of graph is a histogram.



A **histogram** is a visual display of numerical data using bars along a number line with no spaces between the bars. The height of each bar represents either the frequency or relative frequency of data within that interval.

1. The histogram summarizes the lengths, in feet (ft), of a group of adult female sharks.



- a. With your partner, discuss the labels for the vertical and horizontal axes.

Students do not need to write an explanation here, but a sample response is provided for reference. The horizontal axis represents the length of the female sharks. The vertical axis represents the number of sharks with lengths in the various intervals.



Notice and Wonder

Consider creating a T-chart on the board with the headings “Notice” and “Wonder.” Invite students to point out the observations they notice and to cite the details they wonder about regarding the histogram. Use students’ comments to complete the T-chart.



Discuss the term “frequency” with students. Have students pick a color: red, blue, yellow, or green. Determine how frequently each color is chosen as a way to demonstrate frequency to students. Color-code students’ responses on the board to demonstrate the frequency.

9.5.2 Guided Instruction: Histograms (continued)

- b. How many sharks were between 15.5 ft and 16 ft?

9 sharks

- c. How many total sharks were measured to create the histogram?

50 sharks

2. Each bar of a histogram includes the left-end value, but not the right-end value. For example, the first bar includes sharks that are 13 ft and 13.25 ft, but not 13.5 ft.

- a. Discuss with your partner whether or not you can determine the exact lengths of each shark measured from the histogram. Summarize your discussion.

Answers will vary. The exact length of the sharks cannot be determined because the histogram displays the number of (frequency) of sharks for each interval of lengths.

- b. Discuss with your partner your observations about the width of each bar.

Students do not need to write an explanation here, but a sample response is provided for reference. The width of the bars are equal.

3. If male sharks were added to the sample, hypothesize what would happen to the spread of the data.

Answers will vary. Male sharks are typically smaller than female sharks, so adding the lengths of the male sharks will likely not change the range of the data.



To draw attention to the way in which histograms differ from other graphical representations and present data, ask questions like:

- How does each bar of the histogram show the range of lengths of sharks? How does this differ from showing the exact lengths of sharks?
- What is the mode of the data? Can you determine it using the histogram? Why or why not?
- Where do you think the median would be?

9.5.3 Exploration: Histograms

Component Overview: In this activity, students notice how the data in the table translates to the histogram. Students may work independently or with a partner to complete the activity. Partners may work together on all of the questions or divide the items and then share their answers with their partner.

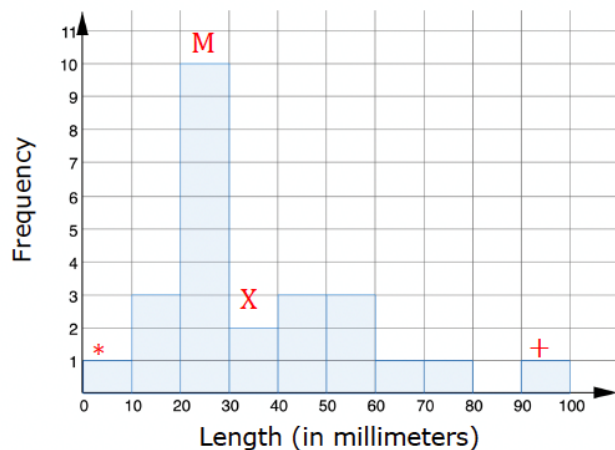


9.5.3 Exploration: Histograms

An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer recorded the earthworms' lengths, in millimeters, in a table and then created a histogram to display the data.

Earthworm Lengths in Millimeters (mm)									
6	11	18	19	20	23	23	25	25	26
27	27	28	29	32	33	41	42	48	52
54	59	60	77	93					



Some students may struggle to track the data with their eyes. Encourage students to color-code different bars in the histogram and to draw a horizontal line from the top of each bar to its corresponding frequency on the y-axis.

1. How many earthworms were measured in the data?
25

2. The number of earthworms measured can be

determined from

- ☐ the data set.
- ☐ the histogram.
- ☒ the data set and the histogram.

9.5.3 Exploration: Histograms (continued)

3. The length of the shortest earthworm can be

determined from

- ☒ the data set.
- ☐ the histogram.
- ☐ the data set and the histogram.

4. Place a "*" above the bar that contains the value for the shortest earthworm.

The smallest data value is in the first column.

5. Determine the length of the longest earthworm. Then, place a "+" above the bar that contains the value for the longest earthworm.

The longest earthworm was 93 mm and is located in the last column.

6. The mean length of the earthworms can be determined

from

- ☒ the data set.
- ☐ the histogram.
- ☐ the data set and the histogram.

7. What is the mean length of the earthworms?

35.92

8. Half of the earthworms are longer than 28 millimeters.

9. Mark an "X" above the bar that contains the mean and mark an "M" above the bar that contains the median.

Added to original histogram

10. Discuss with your partner how the histogram would change if a 100-millimeter long earthworm was added to the data set. Summarize your discussion.

Answers will vary. Adding a 100-millimeter long earthworm to the data set will add another bar to the histogram, since that earthworm would be included in the interval 100 – 110.



To help students discern the differences and make connections between the different ways to display data, ask questions like:

- How does the data set help you visually understand the data?
- Do you think the data is easier to analyze listed individually or grouped in the histogram? Why?
- What are the strengths of just using the data set?
- What are the strengths of just using the histogram?



Challenge students to determine the mode and the range of the data. If students feel comfortable with box plots, have them create a box plot from the data.

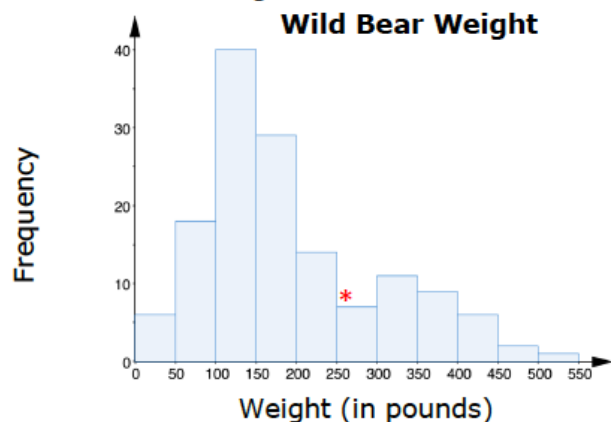
9.5.4 Try It: Analyzing Histograms

Component Overview: Students use a histogram to analyze the data contained in it. Some students will feel comfortable completing this activity independently, while other students will feel more comfortable with support.



9.5.4 Try It: Analyzing Histograms

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data were used to make the histogram shown.



- a. Could this sample contain a bear weighing 250 lbs? Explain.
Yes, the 250 pounds bear will fall between the 250 and 300 weight range.
- b. Mark a "*" above that bar that would contain a 250 –lb bear. *Shown on histogram.*
- c. Based on the graph, what can be concluded about the weight of the heaviest bear?
The heaviest bear is in the last interval of weights, 500 to 550, but does not include 550.
- d. What is a typical weight for the bears in this graph? Include units.
Answers will vary. Between 150 and 200 pounds.
- e. Do more than half of the bears in the study weigh less than 250 lbs?
Yes, more than half of the bears weigh less than 250 pounds.



Invite students to share any initial impressions they have of the histogram regarding the distribution of the data. Remind students to read the introductory paragraph and think about any suppositions they have about the relation of a bear's age to its weight.



9.6 Describing and Interpreting Histograms - Part 2 will focus on student understanding of the shape of histograms and how to detect any gaps, clusters, or outliers in the data set. Position students to begin to explicitly analyze the distribution of data:

- How would you describe the shape of this histogram? Does it look the same on either side of the median? Why or why not?
- How can you use the histogram to determine a "typical" weight?
- Do any of the data points on the histogram seem out of place? Why?

9.5.5 Your Turn!

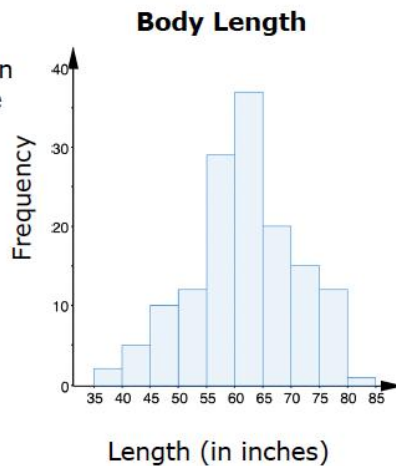
Component Overview: In this activity, students will synthesize what they have learned about analyzing a histogram. Students may complete the activity independently, with a partner, or in a small group.



9.5.5 Your Turn!

1. The histogram summarizes the data on the body lengths of the same 143 wild bears.

- a. Does this sample contain a bear that is 85 inches long? Explain your reasoning.



No, this sample does not contain a bear that is 85 inches long. The largest bear is between 80 and just up to 85 inches.

- b. What is a typical length for the bears in this sample? Include units.
Between 60 and 65 inches.
- c. Is the spread larger for the weight of bears or the length of bears?
The spread is larger for the weight of bears.



Students may struggle to determine the typical length of the bears. Ask questions to further scaffold this section:

- How can you determine which length is typical using a histogram? How can you use the visual display to determine a typical weight?
- How do you determine frequency values between the marked numbers on the y-axis?
- What is the range of the bears' body lengths?

9.5.6 Wrap-Up: Reflection

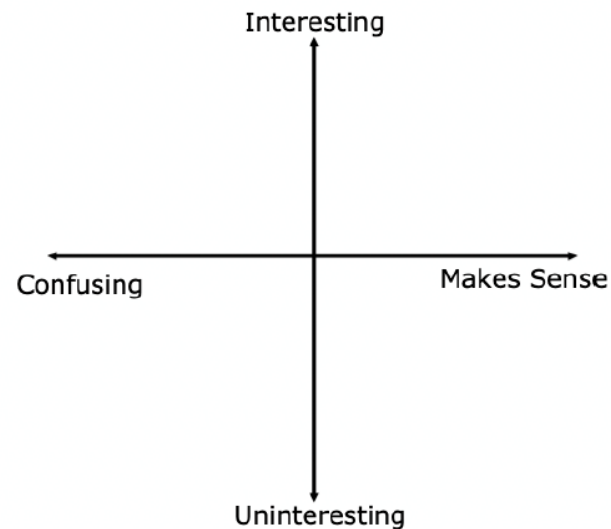
Component Overview: Students assess their understanding of describing and interpreting a histogram within a real-world context. Consider using this reflection to inform partner pairs or small groups for the next lesson, *Describing and Interpreting Histograms - Part 2*.



9.5.6 Wrap-Up: Reflection

- Plot a point in one of the quadrants below to indicate your level of understanding of today's learning target: "I can qualitatively describe and interpret a histogram within a real-world context."

Answers will vary.



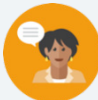
Consider creating a chart to collect and organize data on student responses to the reflection. Put a mark beside each student according to their response. In this way, you will be able to gain better insight on the reflection by analyzing student data as a whole.

	Interesting	Uninteresting	Confusing	Makes Sense
Student 1	X		X	
Student 2		X	X	

9.6 Describing and Interpreting Histograms – Part 2

Lesson Introduction

 Benchmark	MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.		 Learning Targets	- I can describe the shape of a histogram. - I can describe the spread of a data set given a histogram. - I can describe gaps, outliers, and clusters in a histogram.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.	
 Essential Question	- How can you describe the distribution of a set of data on a histogram? - What does a peak indicate? What about a gap? - How do you determine a cluster? An outlier?			
 Lesson Narrative	This lesson is Part 2 of 2 of interpreting histograms. Students begin by applying what they learned in Part 1 to describe the shape of a histogram and consider whether the shape is symmetrical or asymmetrical after reviewing the definitions. Students then examine several histograms and describe their symmetry, skew, and distribution. Students deepen understanding by analyzing the implications of any clusters, gaps, or outliers shown in a histogram.			
 Component Summary	9.6.1 Warm-Up (~5 mins)	Error analysis to determine a “math fail” (SE p. 69)	9.6.4 Guided Instruction (10 mins)	Define and describe the features of a histogram (SE p. 74)
	9.6.2 Guided Instruction (10 mins)	Analyze the shape of a histogram to determine symmetry and skew (SE p. 70)	9.6.5 Collaboration (5 mins)	Analyze the features of histograms to interpret and compare data sets (SE p. 75)
	9.6.3 Guided Practice (10 mins)	Describe the distribution of data in a histogram (SE p. 72)	9.6.6 Your Turn! (5 mins)	Analyze the shapes of two histograms to compare data (SE p. 77)
	9.6.7 Wrap-Up (~5 mins)	Self-assessment of describing and analyzing histograms and their features (SE p. 77)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> - Right-skewed - Left-skewed - Cluster - Gap - Outlier <u>Additional Terms:</u> N/A	(Optional) Colored pencils (Optional) Highlighters	N/A	

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may struggle to understand why the data on each side of a line of symmetry may not be exactly symmetrical. - Students may confuse “left-skewed” and “right-skewed.” - Students may struggle to determine if an interval refers to a distance on the x- or the y-axis.		- Have students consider if the data on each side of a line of symmetry is generally symmetrical as opposed to exact. - Encourage students to describe the distribution of data based on initial visual impressions instead of exact mathematical calculations. - Have students notice the context in which terms, such as “interval,” are used. - If time is an issue, then consider assigning Your Turn! for homework.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Take Turns 9.6.5 Collaboration is an opportunity for students to engage in a Take Turns routine. This encourages students to work independently and then engage in mathematical discourse with a partner to explain and justify their responses. This is particularly effective in deepening student understanding and strengthening analytical skills.		MA.K12.MTR.5.1: Patterns and Structures Students will be working to describe the shape of a histogram and its data set. Clarifications within this standard emphasize recognizing patterns and applying these patterns to mathematical concepts. Utilize the progression of the lesson to highlight the process by which students use details within a problem to form generalizations about a data set.	
 Differentiate Instruction	The display of data through the graphical representations on histograms provides visual entry points for students to engage with the content and begin to analyze the information presented. Additional visual entry points can be accessed through the use of colored pencils to label the features of the histograms to better visualize the story of the data.			
Lesson Notes:				

9.6.1 Warm-Up: Math Fail

Component Overview: Students use an open-ended prompt to examine a retail sign. This activity encourages students to be discerning observers of real-life math.



9.6.1 Warm-Up: Math Fail

1. Discuss with your partner what you notice about this picture.



The price being advertised is actually more than the original price.



Consider encouraging students to think about how this sign came to be. Allow students to demonstrate creativity with their answers. Use sentence stems to encourage conversation such as “I think...” or “I wonder...”

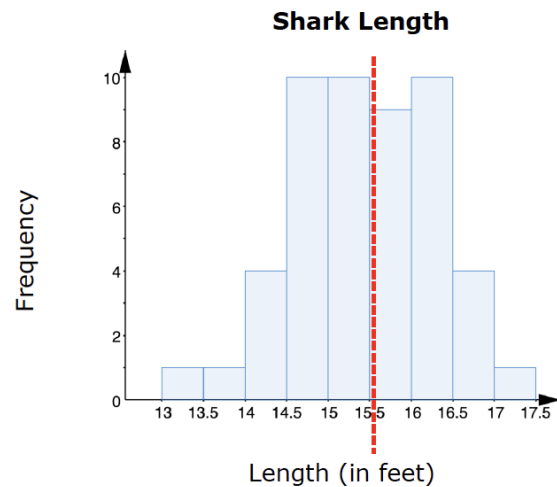
9.6.2 Guided Instruction: Shape of Distributions

Component Overview: Students describe the shape of a distribution displayed in a histogram. Students consider if a histogram is symmetric or asymmetric as well as if it skews to the right or to the left. Encourage students to consider how the shape of a distribution allows them to make generalizations or draw conclusions about a data set.



9.6.2 Guided Instruction: Shape of Distributions

1. Consider the histogram of the lengths of female sharks.



- a. Sketch the line of symmetry for the histogram.
Answers will vary.
- b. If the histogram is folded along the line of symmetry, will each half line up perfectly?
No, the histogram will not line up perfectly.
- c. Do you think all histograms will be symmetrical?
Answers will vary. No, I do not think all histograms can be described as symmetrical.



Have students compare and contrast the exact symmetry of the heart with the general symmetry of the histogram. Encourage students to describe if they think that the term “symmetrical” can still be used for things that are not exactly symmetrical. Challenge students to think of real-life items that can be generally, but not exactly, symmetrical.

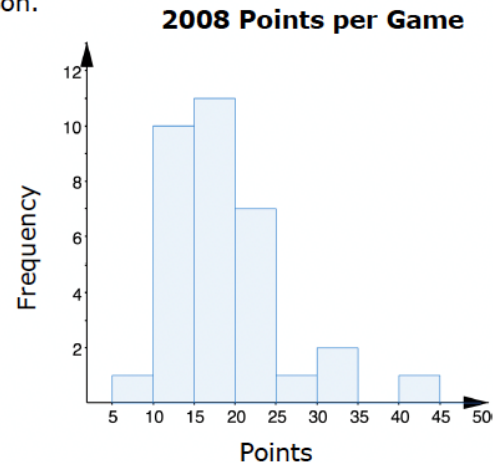


This section provides visual entry points through the graphical display of data on the histogram. Auditory entry points are provided through partner discussion. Consider additional kinesthetic entry points to engage all learners.

9.6.2 Guided Instruction: Shape of Distributions (continued)

The shape of the shark lengths histogram can be described as **symmetric**, but histograms can also have other shapes.

- The histogram displays the number of points per game scored by a college basketball player during the 2008 season.



Discuss with your partner what you notice about the shape of the distribution of points scored per game.

Students do not need to write an explanation here, but a sample response is provided for reference. Most of the data in this set is on the left side of the histogram. There are fewer large values in this data set compared to small values.

A **right-skewed** graph has a longer trail of data to the right and the peak of the data on the left.



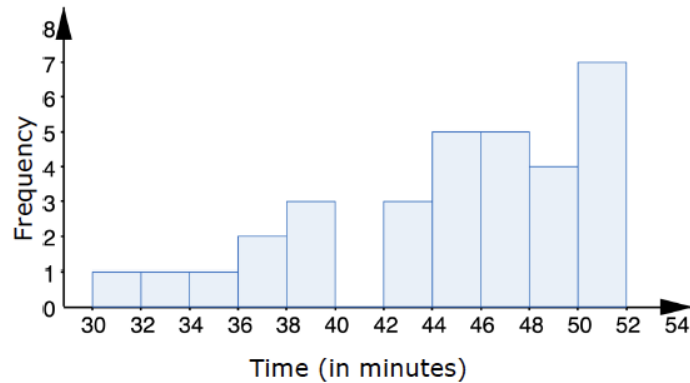
Synthesize student learning through the use of intentional questioning:

- Many words have the prefix “a-”: asocial, anonymous, apolitical, asymmetric. What does this prefix mean?
- How does it help you understand the meaning of “asymmetric”?
- How would you describe the appearance of the data in the histograms without using the words “symmetric” or “asymmetric”?

9.6.2 Guided Instruction: Shape of Distributions(continued)

3. The time to complete a unit exam for 32 students is shown on the histogram.

Time to Complete a Unit Exam



Write an observation about the shape of the distribution of time spent to complete the exam.

Answers will vary. Most of the data in this data set is on the right side of the histogram with less data trailing on the left side.

A **left-skewed** graph has a longer trail of data to the left and the peak of the data on the right.



Ask students what they think the word “skew” means outside of a mathematical context. Encourage students to think of examples when they might have heard the word “skew” and brainstorm contexts in which they could use the word in their everyday lives. Connect this discussion to the description of histograms as either being left-skewed or right-skewed.

9.6.3 Guided Practice: Describing Shape of Distributions

Component Overview: Students describe the shape of the distribution of histograms. Students use the distribution of a histogram to make inferences and draw conclusions about a data set. Encourage students to consider how the data reflects real-life situations.



9.6.3 Guided Practice: Describing Shape of Distributions

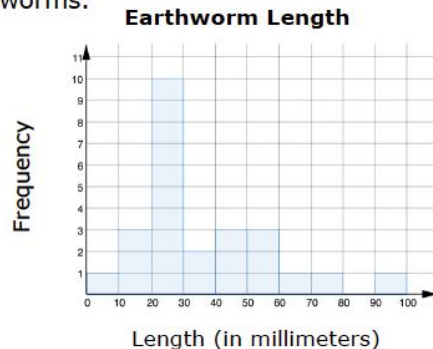
- This histogram shows the ages of students in a swimming class.



The shape of the distribution of ages can be best described as

- ☐ symmetric.
- ☐ right skewed.
- ☒ left skewed.

- The histogram shows the length, in millimeters, of 25 earthworms.



The shape of the distribution of the earthworms' lengths can be best described as

- ☐ symmetric.
- ☒ right skewed.
- ☐ left skewed.



This section is intentionally scaffolded through the use of multiple choice to describe the skew of histograms. Encourage students to refer to the previous section if they are having trouble determining the direction of the skew.



To provide additional scaffolding, encourage students to draw a vertical line down the center of the histogram and describe what they see on each side. This will provide a visual for students to connect to their descriptions.

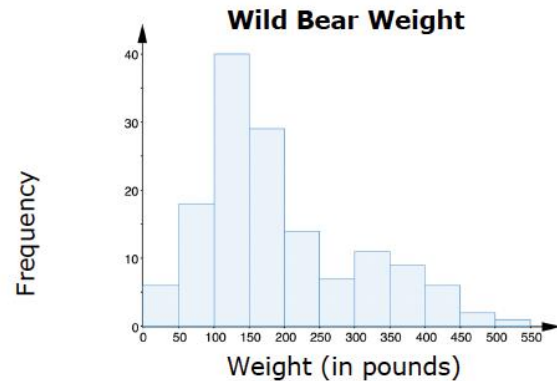


Provide additional scaffolding by asking guiding questions like:

- Where is the center of the histogram? Are there more data points to the left or right of it?
- Where is the peak of the histogram? What does this tell you about skew?

9.6.3 Guided Practice: Describing Shape of Distributions (continued)

3. In one study of wild bears, researchers measured the weights, in pounds, of 143 wild bears that ranged in age from newborn to 15 years old. The data were used to make this histogram.



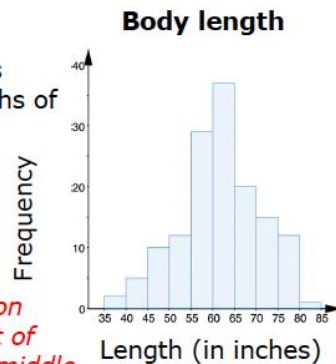
Describe the shape of the graph given the context.

The shape of the Wild Bear Weight histogram is right-skewed. Most of the bears weighed between 50 and 250 pounds, but the range goes all the way up to almost 550 pounds.

4. The histogram summarizes the data on the body lengths of the same 143 wild bears.

Describe the shape of the distribution of the lengths of bears.

The shape of the distribution is roughly symmetric. Most of the data values are in the middle of the distribution.



Consider having students use colored pencils or highlighters to mark or note areas of interest on the histograms. Some students may also benefit from drawing a horizontal line from the top of each bar to its corresponding value on the vertical axis.

9.6.4 Guided Instruction: Features of Histograms

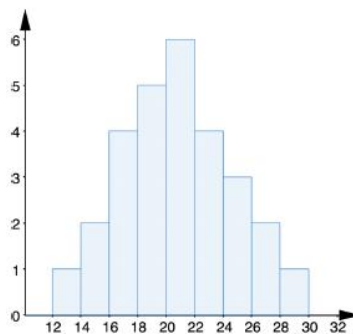
Component Overview: Students consider the features of histograms. Students use the terms “peak”, “cluster”, “gap,” and “outlier” to describe a histogram and make inferences about the data set represented.



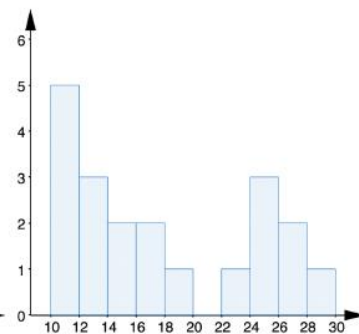
9.6.4 Guided Instruction: Features of Histograms

Below are two histograms with different shapes and features.

Histogram A



Histogram B



1. Histogram A is ☐ symmetrical and has a peak near 20.
☐ not symmetrical

2. Histogram B is ☐ symmetrical and has
☐ not symmetrical

- ☐ one peak.
☐ more than one peak.



A **cluster** is formed when data points are in a close group or numerical data have similar values on a number line.

3. ☐ Histogram A
☐ Histogram B has two clusters.



Invite students to consider when they have heard or used each word—cluster, gap, peak, outlier—in their daily lives. With a partner, have students discuss the everyday meanings and the mathematical meanings of the terms. Encourage students to note any overlap in the meanings.



Have students fold a piece of paper into four equal sections and then label each section with one of the following terms: cluster, gap, peak, and outlier. In the corresponding section, have students write the mathematical definition of the term and draw an illustration depicting it.

9.6.4 Guided Instruction: Features of Histograms (continued)

A **gap** shows a location with no data values. On a histogram, a space between the bins indicates a gap, or a frequency of zero for that bin.

4. ☐ Histogram A
☒ Histogram B has a gap between 20 and 22.



An **outlier** is a value that is much higher or much lower than the other values in a data set.

5. If the data value ☐ 12
☐ 24
☒ 40 were added to Histogram A, it would be considered an outlier.



To encourage conceptual understanding over memorization, ask questions like:

- Why do you think outliers occur? What would be an outlier in a histogram showing the heights of students in our class?
- How could you determine that a number is an outlier? Are outliers dependent on the situations they depict? Why or why not?

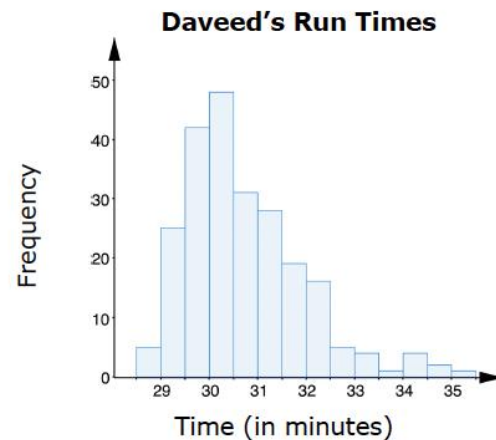
9.6.5 Collaboration: Features of Histograms

Component Overview: Students can summarize or synthesize their findings in this activity. Partners will work together on all of the items or each solve half of the problems and then share their answers. Also consider placing students in small groups, with each student solving a problem and then using a jigsaw method to share their findings.



9.6.5 Collaboration: Features of Histograms

1. Daveed has been tracking the time, in minutes, it takes him to run 4 miles for the last 10 years. A histogram of Daveed's run times is shown.



Take Turns

Consider using a Take Turns routine for the Collaboration. Students will take turns answering each question. The student who does not answer a particular question will explain their partner's answer in their own words. If a partner answers or explains inaccurately, the pair can then work together to solve the problem.

- a. The shape of run times can be best described

as ☒ right skewed.

- b. Daveed's typical run time was about 30.5

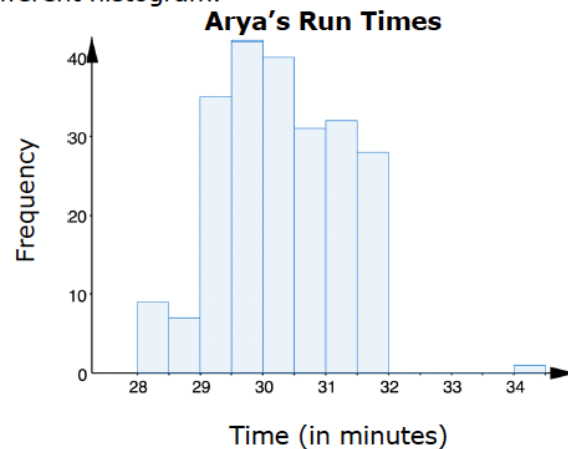
minutes. Most of his run times were between

29.5 to 31 minutes. Not many of his runs

were ☒ longer than 32.5 minutes.

9.6.5 Collaboration: Features of Histograms (continued)

2. Arya used the data of her 4-mile run times to create a different histogram.



- a. Discuss with your partner what you notice about the interval widths. Determine if there are any unusual features including clusters, gaps, or outliers. Write a detailed description of your discussion.

Answers will vary. The x-axis begins at 28 minutes and then increases by 0.5. There is a gap in Arya's run times from 32 to 34. The run time between 34 and 34.5 minutes could be an outlier since it is not close to the other data values. Most of Arya's runs last between 29 to 31.5 minutes.

- b. At what interval in the histogram of Arya's times is there an outlier?

Between 34 to 34.5 minutes.

- c. Would this same data value also be considered an outlier in the histogram of Daveed's times?

No, Daveed's has runs that are between 34 to 35 minutes, and runs that are even longer.

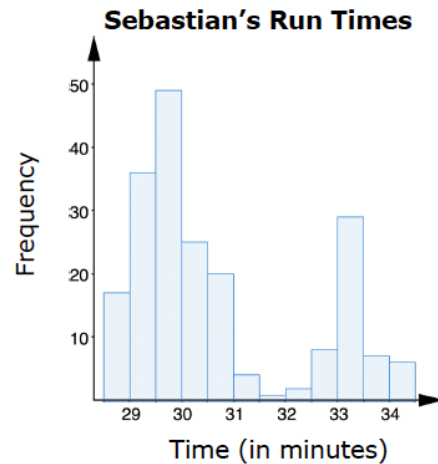


To guide student discussion and provide scaffolding support for students who may struggle with the partner discussion, ask guiding questions like:

- *What do you notice about the intervals? What does this tell you about the data set?*
- *Are there any gaps in the data? What does this mean for outliers?*
- *How can you describe Arya's run times? What data backs up this statement?*

9.6.5 Collaboration: Features of Histograms (continued)

3. Sebastian used the data of his 4-mile run times to create a third histogram.



- a. Which feature is depicted in Sebastian's histogram?
Answers will vary. Sebastian has two clusters of data values, with two peaks. One peak is between 29.5 and 30 and the other is between 33 and 33.5.
- b. Describe the spread and shape of the distribution of Sebastian's run times.
Answers will vary. Sebastian's run times has two peaks and the data took a dip between 31 and 32.5 minutes. Perhaps the first cluster of runs are at a time of day that he has more energy and the second set is at a time when he is more tired.



MA.K12.MTR 5.1: Patterns and Structures
 Invite students to recognize patterns in the histograms in this activity and then draw inferences or form generalizations about the individuals running based on the histograms.



Allow students to verbally explain their answers to the questions. Some students struggle to convey their words in writing but can offer detailed verbal observations. Both written and verbal answers are valid.

9.6.5 Collaboration: Features of Histograms (continued)

4. Different data will change the intervals for the horizontal axis and will affect the histogram and its features: symmetry, skewness, gaps, clusters, outliers, and the range. With your partner, review all three run-time scenarios and discuss the differences between the histograms.

Answers will vary. Students do not need to write an explanation here, but a sample response is provided for reference.

Daveed: This data set is right skewed. Daveed's run times range from 28.5 to 35.5 minutes.

Arya: This data set has a gap on the right side on the graph from 32 to 34 minutes. Her run that took more than 34 minutes looks like an outlier since no other runs took that long.

Sebastian: This data set has two peaks. He has a low frequency between 31.5 to 32 minutes.



Students may struggle to compare how the data sets impact the features of the histograms. If you notice partner pairs struggling to engage in discourse, ask questions to provide scaffolding:

- *How are the histograms representing Daveed's, Arya's, and Sebastian's run times similarly? How do they differ?*
- *What summary statement could you make about Daveed's run times? Arya's? Sebastian's?*

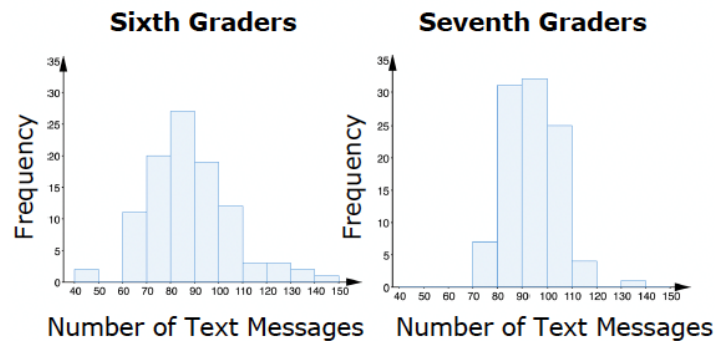
9.6.6 Your Turn!

Component Overview: In this Your Turn!, students compare text message data from 6th and 7th graders using histograms. Students analyze the histograms to describe their shape and determine which class has greater spread and which class is more alike in their cell phone usage.



9.6.6 Your Turn!

- The histograms show the number of text messages sent in one week by a group of 100 sixth-grade and 100 seventh-grade students.



If you notice students struggling to complete this section independently, ask questions to further scaffold:

- Would you describe these histograms as symmetrical or asymmetrical? Why? What does that tell you about the data?
- How can you determine which grade has the largest spread?
- Will the class with the largest spread be more or less alike in their cell phone usage? Why?

- Describe the shape of the histogram for the number of text messages sent in one week by sixth-grade students.

Answers will vary. The sixth-grade data is roughly symmetric.

- Which group of students has the larger spread?

Sixth-Grade Students.

- Explain if sixth graders or seventh graders tend to be more alike in terms of the number of text messages they send.

Answers will vary. Seventh graders are more consistent because the data has a smaller range and the data values are closer together.

9.6.7 Wrap-Up: Reflection

Component Overview: Students reflect on how they feel about describing the shape of a histogram using mathematical language. Consider using this data to inform misconceptions to address in a following lesson or prioritize students to check in with before moving forward with new content.



9.6.7 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of parts of analyzing a histogram. Then provide a brief explanation or evidence to support your self-assessment.

Answers will vary.

1. I can describe the shape of a histogram.

I need help.
I can't get started. I'm getting there. I understand
and have evidence.



2. I can describe gaps, outliers, and clusters in a histogram.

I need help.
I can't get started. I'm getting there. I understand
and have evidence.



3. I can describe the spread of a data set given a histogram.

I need help.
I can't get started. I'm getting there. I understand
and have evidence.



Encourage students to explain their responses in as much detail as possible using specific examples from the lesson. If needed, allow students to use sentence stems such as:

- “I feel comfortable with...”
- “One question I still have is...”
- “An activity that I struggled with was...”

9.7 Describing and Interpreting Line Plots

Lesson Introduction

 Benchmark	MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.		 Learning Targets	- I can describe and interpret data represented on a line plot. - I can determine the similarities and differences between a line plot and a histogram.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.	
 Essential Questions	- How can you read a line plot? - How can you analyze a line plot? - What are the similarities between a line plot and a histogram? - What are the differences between a line plot and a histogram?			
 Lesson Narrative	In this lesson, students review the definition of line plots and analyze line plots to answer questions about data sets. Students then compare line plots and histograms, considering the way data is displayed in each and similarities and differences between the two graphical representations. Students then consider the information they can gather from each data display and apply that knowledge to answer questions about the data.			
 Component Summary	9.7.1 Warm-Up (~5 mins)	Interpret data on a histogram (<i>SE p. 78</i>)	9.7.3 Exploration (10-15 mins)	Compare line plots and histograms (<i>SE p. 80</i>)
	9.7.2 Guided Instruction (5-10 mins)	Analyze data on a line plot (<i>SE p. 79</i>)	9.7.4 Your Turn! (5-10 mins)	Demonstrate understanding of how to read and compare line plots and histograms (<i>SE p. 81</i>)
	9.7.5 Wrap-Up (~5 mins)	Reflect on understanding of reading and comparing line plots and histograms (<i>SE p. 83</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Line plot <u>Additional Terms:</u> - Histogram - Range		(Optional) Colored pencils	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse/switch which is the line plot and which is the histogram. - Students may incorrectly state that the range of a line plot or histogram is the highest number on the number line used to create the display. - When asked how many in a category, students may list the number labeled on the line and not the number of dots.	- Refresh student understanding of range as the difference between the maximum and the minimum value in the data set, if needed, during the 9.7.2 Guided Instruction, 9.7.3 Exploration, and 9.7.4 Your Turn!. - Consider providing sentence stems and scaffolding prompts for the open-ended questions throughout the lesson. - Consider projecting the line plots and histograms on the board for those who are visually impaired.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 9.7.4 Your Turn! Provides an opportunity for students to engage in a Compare and Connect routine to draw connections between line plots and histograms. This routine encourages meta-awareness and mathematical discussion through partner talk.	MA.K12.MTR.2.1: Multiple Representations The line plot and histogram show different ways the weight of the dogs can be represented. These will be used for comparisons of the data to see which graph is the best to use for different purposes.
 Differentiate Instruction	Notice 9.7.2 Guided Instruction offers visual and auditory entry points for students to engage with the content. Graphical displays of data help students engage visually. Consider additional visual entry points, such as using colored pencils, to mark up the graphs. The analysis of the data is conducted through partner discussion, providing additional auditory entry points as students verbalize their thinking and listen to their partners' analysis.	
Lesson Notes:		

9.7.1 Warm-Up: Gathering Data

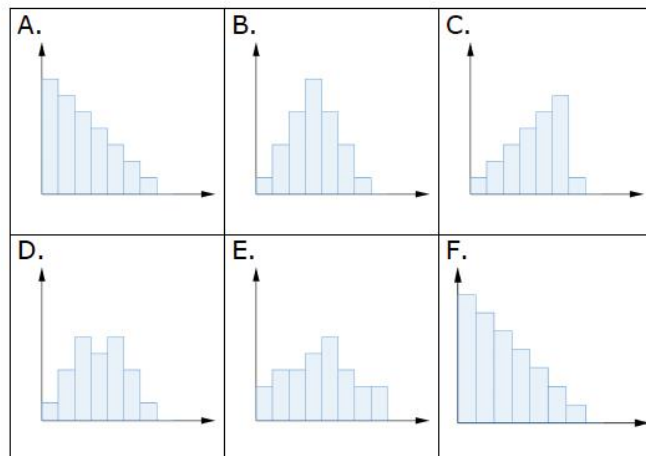
Component Overview: Students categorize histograms into three categories: symmetric, skewed left, and skewed right. This will help students get ready to interpret data on the line plots in later activities.



9.7.1 Warm-Up: Gathering Data

1. Categorize each histogram according to its shape by placing the letter in the appropriate box in the chart below.

Symmetric	Skewed Left	Skewed Right
<i>B, D, E</i>	<i>C</i>	<i>A, F</i>



If students are struggling to categorize the histograms, encourage them to draw lines down the center of the histograms to compare each side.

9.7.2 Guided Instruction: Describing and Interpreting Data Represented on Line Plots

Component Overview: In this Guided Instruction, students will analyze line plots. Students will analyze data in two different line plots. Spend time in the beginning letting students understand how to read a line plot through the questioning provided to alleviate confusion later.



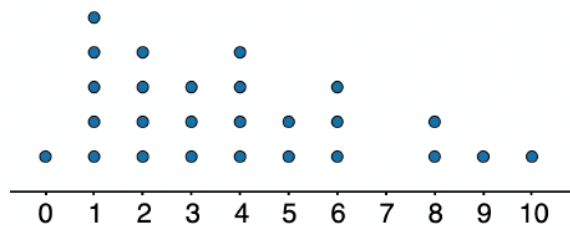
9.7.2 Guided Instruction: Describing and Interpreting Data Represented on Line Plots

Line plots are another way to display numerical data.



A **line plot** is a visual display of data values where each data value is shown as a dot or mark above a number line. This is also known as a dot plot.

- The line plot shows the number of hours per week that students reported spending on homework.



Hours Spent on Homework per Week

- How many students were surveyed?
26 students
- How many students reported spending 1 hour on homework each week?
5 students
- What percentage of the students reported spending 4 hours or fewer on homework each week?
65.38%
- What is the range of time spent on homework in this group?
10 hours



Students commonly misread line plots because they do not understand each dot represents a separate data point. Ensure students understand how to read line plots by asking questions like:

- What does each dot represent?
- Which amount of time did the most students respond with? How do you know?
- What was the least popular answer? How do you know?

9.7.2 Guided Instruction: Describing and Interpreting Data Represented on Line Plots (continued)

With your partner, discuss and summarize your discussion on the following questions about the hours spent on homework per week.



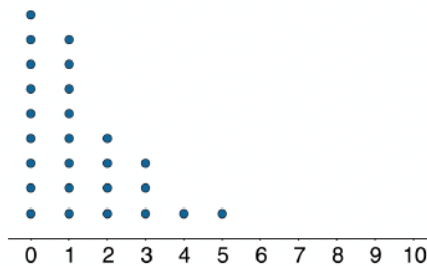
Think-Pair-Share

For this activity, allow students a moment to think of their own answers before sharing with their partner. Listen for different student answers as they discuss with their partner and pick opposing answers that could both work as the answer to share with the class. Encourage discourse amongst the class to determine why there are multiple answers for each question.

- e. Explain whether or not the mean time spent on homework by this group can be determined.
Answers will vary. The mean can be determined by dividing the total sum of the hours spent on homework by the number of students surveyed. The line plot displays the frequency of each time so we are able to add the values together to find the sum.
- f. Someone said, "In general, these students spend roughly the same number of hours doing homework." Do you and your partner agree? Explain your reasoning.
Answers will vary.
- g. Describe the shape and special features of the distribution of hours spent on homework. Find another pair of partners and compare your descriptions.
Answers will vary. The shape of the distribution is slightly skewed right. There does not seem to be any outliers but there is a gap in the data at 7 hours.

9.7.2 Guided Instruction: Describing and Interpreting Data Represented on Line Plots (continued)

2. The same students were asked to estimate how many hours a week they spend talking on the phone. This line plot represents their reported number of hours of phone usage per week.



Hours on the Phone per Week

- a. Describe the spread of the data.

Answers will vary. Most of the values are close together and are small values, meaning that the students don't spend much time talking on the phone.

- b. Overall, are these students more alike in the amount of time they spend talking on the phone or in the amount of time they spend on homework? Explain your reasoning.

Answers will vary. These students are more alike in the amount of time they spent talking on the phone because all of the data is clustered to the left and none of the students spend more than 5 hours on the phone per week.

- c. Compare your reasoning with your partner. Make adjustments to your reasoning after the discussion with your partner, if necessary.



Encourage students to use the information from the Warm-Up to help answer question 2a. For question 2b, position students to discuss what “being alike” looks like in a line plot through questions like:

- How would you describe each line plot?
- Which line plot has more spread? What does that tell you?
- Does the line plot with more or less spread show whether students are more alike? Why?

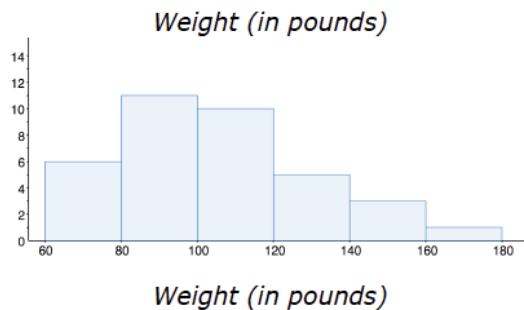
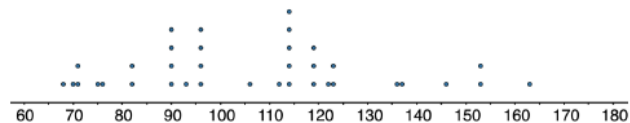
9.7.3 Exploration: Comparing Displays

Component Overview: In this Exploration activity, students will compare a dot plot and a histogram to determine which graphical representation is best suited to represent certain types of data and why.



9.7.3 Exploration: Comparing Displays

- The dot plot and histogram display the weights, in pounds, of 36 dogs at a dog show.



- ☐ The line plot
☐ The histogram
☐ Both the line plot and histogram

 can be used to determine the number of dogs that weigh 90 lbs.
- How many dogs have a weight of 90 lbs?
4
- Find the range of the dogs' weights.
95
- With your partner, discuss what you notice about the shape of the distribution.
The shape of the distribution is skewed right.
- Which display was more useful when describing the shape?
Answers will vary.



MA.K12.MTR 2.1: Multiple Representations
The dot plot and histogram show different ways the weight of the dogs can be represented. These will be used for comparisons of the data to see which graph is the best to use for different purposes.



Encourage students to label the dot plot and the histogram to cut down on confusion. Consider providing students with sentence stems for question 1d and 1e to scaffold student response.

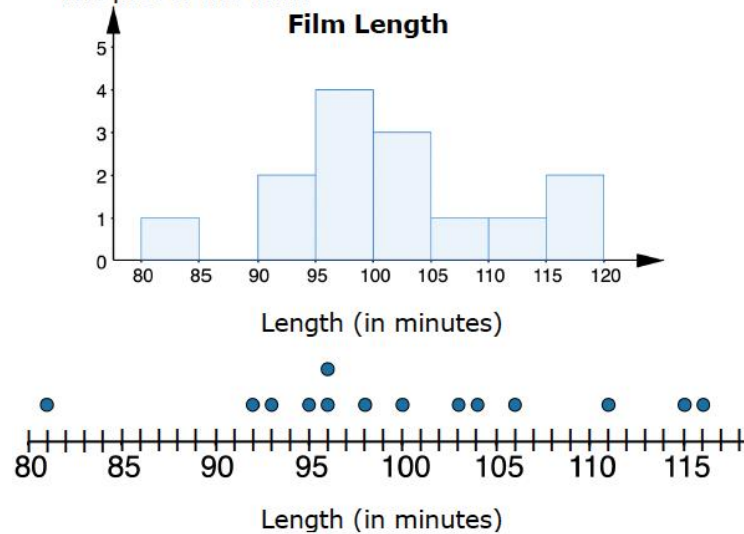
9.7.4 Your Turn!

Component Overview: Students analyze data presented in both a histogram and line plot to answer questions about a data set. Students then consider which representation can be used to determine the range of a data set and why. Students then analyze another line plot and describe the shape and distribution of the data presented.



9.7.4 Your Turn!

1. A movie critic recorded the length, in minutes, of a sample of Pixar films and then made a histogram and a line plot of the data.



- a. How many film lengths are recorded? **14**
- b. How many films are between 95 minutes and 100 minutes in the histogram? **4**
- c. How many films are between 95 and 100 minutes in the line plot? **5**
- d. What is the value of the range? **35**
- e. Discuss with your partner which graph was used to determine the range and why.

The line plot was used because the line plot shows the individual data values so the minimum and maximum can be determined.



Compare and Connect

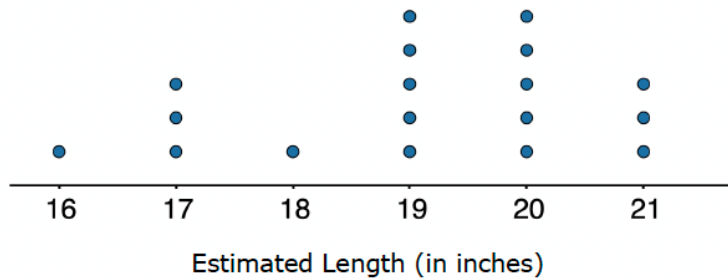
Help students compare and make connections between the histogram and line plot. Model thinking out loud to guide student thinking by asking questions, wondering how the graphs are similar, and considering the information that can be gathered from each.



Consider having students create a Venn diagram to show the similarities and differences between a histogram and a dot plot. This provides a visual entry point for learners to discern the differences between the two representations and draw conclusions about which graph is better suited to provide information about different data sets.

9.7.4 Your Turn! (continued)

2. A teacher drew a line segment that was 20 inches long on the blackboard. She asked each of her students to estimate the length of the segment and used their estimates to draw this line plot.



- a. How many students were in the class?

18

- b. Were students generally accurate in their estimates of the length of the line?

Answers will vary. Students were generally accurate because most of the students estimated the distance to be close to the actual distance, 20.

- c. Describe the shape and distribution of the estimated lengths.

The shape and distribution of the lengths are skewed left.



Students may struggle to determine whether or not the students were accurate in their estimates. Ask guiding questions to help students draw conclusions based on the data:

- Where do most of the data points on the line plot fall?*
- Is this close to the length of the actual line? What does that tell you about the accuracy of the estimates?*



Challenge early finishers to create a histogram using the data from the line plot. Encourage students to explain which graphical representation of the data they prefer to use and why.

9.7.5 Wrap-Up: Reflection

Component Overview: Students reflect on the lesson and choose statements to describe their understanding of the content.



9.7.5 Wrap-Up: Reflection

1. Check all the statements that apply to your understanding.

Answers will vary.

- ☐ I can read a line plot.
- ☐ I need help reading a line plot.
- ☐ I can describe and analyze data represented on a line plot.
- ☐ I need help describing and analyzing data represented on a line plot.
- ☐ I can determine the similarities and differences between a line plot and a histogram.
- ☐ I need help to determine the similarities and differences between a line plot and a histogram for the same data.

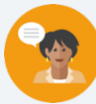


Consider using this self-reflection to inform small groups for remediation or partner-pairs for upcoming lessons.

9.8 Comparing Distributions

Lesson Introduction

 Benchmarks	MA.6.DP.1.2 Given a numerical data set within a real-world context, find and interpret mean, median, mode, and range. MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.		 Learning Targets	- I can find and interpret the mean and median of data sets. - I can use mean and median to compare data sets. - I can compare the distribution of data sets.	
 Benchmark Coherence	Building On		Working Toward		
	MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.		
 Essential Questions	- When would it be better to use a line plot over a histogram? - What does the mean tell you about a data set? - What does the distribution tell you about a data set?				
 Lesson Narrative	In this lesson, students will build on their understanding of histograms and line plots. Students will find the mean or median within a data set and compare it to another data set. Students will synthesize this information to use for their reasoning behind their answers. Ensure students have a firm understanding of how to read histograms and line plots before beginning this lesson.				
 Component Summary	9.8.1 Warm-Up (~5 mins)	Engage in “Which one doesn’t belong?” routine (<i>SE p. 84</i>)	9.8.3 Guided Practice (15 mins)	Determine the mean and median in line plots, histograms, and tables (<i>SE p. 85</i>)	
	9.8.2 Collaboration (10 mins)	Match labels to line plots (<i>SE p. 84</i>)	9.8.4 Your Turn! (10 mins)	Use mean and median to compare data sets and compare the distribution of data sets (<i>SE p. 87</i>)	
	9.8.5 Wrap-Up (~5 mins)	Compare distribution sets of histograms (<i>SE p. 89</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Distribution - Mean and Median - Histogram and Line Plot		(Optional) Paper		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly switch the median and the mode. - Student may add numbers that are labels at the bottom instead of the amount in each category. - Students may incorrectly think that the given histograms show exact numbers for the data points.	Make sure students can read each chart and can see all of the labels. Consider projecting each line plot or histogram on the board for those who are visually impaired
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Which One Doesn't Belong? The 9.8.1 Warm-Up engages students in a “Which one doesn't belong?” routine. Use this as an opportunity to encourage critical thinking and reasoning skills. Each of the four options may not belong for some reason. As students discuss their responses, encourage them to justify their thinking using the pictures.	MA.K12.MTR.6.1: Reasonableness Throughout this lesson students make comparisons between data sets before finding measures of central tendency. Students have the opportunity to predict solutions based on the given information, strengthening their ability to reason mathematically about data sets and to consider the reasonableness of their hypotheses.
 Differentiate Instruction	Line plots and histograms provide visual entry points for learners throughout this lesson. Seeing the data displayed graphically can help students make connections and hypothesize about measures of central tendency as well as the distribution and spread of the data, which is critical to being able to compare data sets.	
Lesson Notes:		

9.8.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students engage in a “Which One Doesn't Belong?” routine to determine the license plate that doesn't belong with the rest.



9.8.1 Warm-Up: Which One Doesn't Belong?

- Which license plate doesn't belong? Explain your reasoning.



Answers will vary.



Which One Doesn't Belong?

Engage students in a “Which one doesn't belong?” routine, prompt students to explain their rationale for their choice, and give opportunities for students to refine and clarify their answers.

9.8.2 Collaboration: Comparing Distributions

Component Overview: Students work with a partner to match line plots to their corresponding grade levels based on summaries describing the data for each class.



9.8.2 Collaboration: Comparing Distributions

- Clare recorded the amount of time students in the sixth, eighth, and tenth grades spent doing homework, in hours per week. She made a line plot of the data for each grade and provided the following summary.
 - Students in sixth grade tend to spend less time on homework than students in eighth and tenth grades.
 - The homework times for the tenth-grade students are more alike than the homework times for the eighth-grade students.

Use Clare's summary to match each line plot to the correct grade (sixth, eighth, or tenth).

Grade Level	Line Plot
<i>eighth grade</i>	
<i>sixth grade</i>	
<i>tenth grade</i>	
	Time (in hours)



Consider encouraging students to label the line plots by circling the median, drawing a line down the center, and comparing both sides to draw conclusions about the data. This will provide visual entry points for students to analyze the data presented in the line plots and allow for easier comparison.



Take Turns

With a partner, students take turns deciding which grade belongs with each graph. When Partner A chooses an answer, Partner B will ask them to give reasons for their answer and then agree or disagree with a reason. Students will discuss until they come to an agreement. Consider switching partners or providing prompting questions if students cannot come to an agreement.

9.8.3 Guided Practice: Comparing Distributions

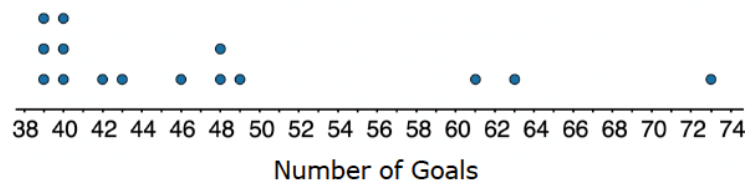
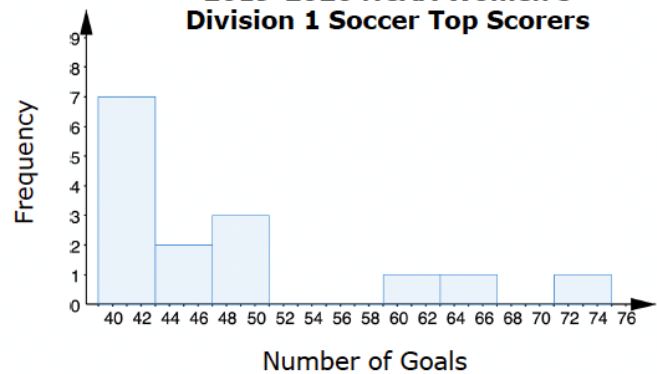
Component Overview: Students analyze a data set presented as a histogram and a line plot and in a table. Students determine which measures of central tendency can be found using a line plot and a histogram and consider which measure of central tendency will be larger given the data set. Students then analyze the same data set presented in a table using only the top ten data points to examine how the measures of central tendency change.



9.8.3 Guided Practice: Comparing Distributions

1. A histogram and line plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season are shown.

2019-2020 NCAA Women's Division 1 Soccer Top Scorers



- a. The

☐ histogram
☒ line plot
☐ histogram and line plot

 can be used to determine the mean.
- b. The

☐ histogram
☒ line plot
☐ histogram and line plot

 can be used to determine the median.



Differentiation is built in throughout this component to provide auditory and visual entry points. Graphical displays of data allow students to visualize the center and spread of the data set. Partner discussion provides auditory entry points.



This section is intentionally scaffolded to solidify student understanding of how data impacts measures of central tendency. Students begin by using visual representations of data to generalize about mean and median, then progress to using data presented in a table to find mean and median. Encourage students to discuss how the different graphical displays impact the measures of center that can be calculated.

9.8.3 Guided Practice: Comparing Distributions (continued)

- c. Which measure of central tendency do you expect to be larger, the mean or the median? Explain your reasoning.

Answers will vary. I expect the mean to be larger because the larger values that cause the distribution to be skewed to the right increase the average.

2. The table lists the data for the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.

Rank	Player	Goals
1	Evelyne Viens, South Fla.	73
2	Catarina Macario, Stanford	63
3	Raimee Sherle, Boise St.	61
4	Ally Watt, Texas A&M	49
5	Tziarra King, NC State	48
5	Deyna Castellanos, Florida St.	48
7	Alexa Genas, Campbell	46
8	Morgan Weaver, Washington St.	43
9	Peyton DePriest, Middle Tenn.	42
10	Elise Flake, BYU	40
10	Samantha Dewey, Xavier	40
10	Hayley Younginer, Wake Forest	40
13	Dakota Mills, Saint Joseph's	39
13	Katie McClure, Kansas	39
13	Taylor Kornieck, Colorado	39

- a. Find the mean number of goals. Interpret the mean.

47.3. The average number of goals scored for the top 15 women during the 2019-2020 NCAA Women's Division 1 soccer season was 47.3 goals.



Students may struggle to predict the measure of center that will be larger. Help students think through and estimate their answers by asking questions like:

- What do you notice about the data? Is there wide variation?*
- Does a wide variation affect mean or median more? Why?*
- Are there any outliers? How do outliers typically affect measures of central tendency?*

9.8.3 Guided Practice: Comparing Distributions (continued)

- b. Find the median number of goals. Interpret the median.

43. The median number of goals scored for the top 15 women during the 2019-2020 NCAA Women's Division 1 soccer season was 43 goals. Half of the top 15 scored fewer goals and half scored more.

- c. Which value is larger, the mean or the median?
mean

- d. How would the mean and median change if you found the mean goals and median goals for the top 8 scorers?

Answers will vary. The mean and the median would both be larger numbers.

- e. Find the mean number of goals for the top 8 scorers.

53.875

- f. Find the median number of goals for the top 8 scorers.

48.5

- g. Discuss with your partner how the measures of central tendency changed. Write your conclusion.

Answers will vary. Both the median and the mean increased since the lowest seven scorers were removed from the data set.



Take Turns

Encourage students to engage in mathematical discourse through a Take Turns routine. Students work in pairs or small groups and take turns in the work of the component. Circulate as students work and encourage partners to question each other's work, justify their own claims, and explain their thinking.



Consider allowing students to explain their answers to questions verbally instead of in writing. Oral explanations may help students who struggle with written response articulate their thinking and provide another valid form of assessment.

9.8.4 Your Turn!

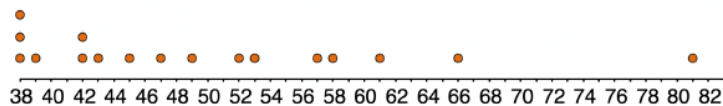
Component Overview: Students compare two data sets displayed on line plots. Students begin by describing the differences between the data sets, then hypothesize about the mean and medians and compare their predictions with a partner. Students then find and interpret the measures of center for one of the data sets by using data displayed in a table.



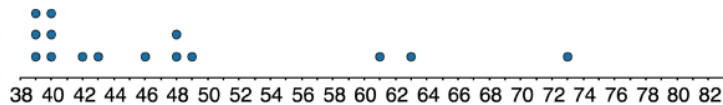
9.8.4 Your Turn!

- The line plots show the top 15 women goal scorers during the 2016-2017 and 2019-2020 NCAA Women's Division 1 soccer season.

2016-2017 Season



2019-2020 Season



Number of Goals

- What is different about the top 15 goal scorers during the 2016-2017 soccer season and the top 15 goal scorers during the 2019-2020 soccer season?

Answers will vary. The number of goals are closer together (less variation) during the 2019-2020 season versus the 2016-2017 season. There is a larger range for the 2016-2017 season.

- Make a hypothesis determining whether the mean of the 2019-2020 season or the mean of the 2016-2017 season is higher. Explain what influenced your hypothesis.

Answers will vary. My hypothesis is that the mean of the 2016-2017 season is higher because that season has more data points with larger values.



Students may struggle to compare the data sets and hypothesize about the measures of center. Provide scaffolding by asking questions like:

- How can you describe the data of the 16-17 season? The 19-20 season?
- Does either season have outliers? How will this impact the data?
- Which measure of center is more impacted by outliers? How can this help you hypothesize about the means and medians of the data sets?

9.8.4 Your Turn! (continued)

- c. Make a hypothesis determining whether the median of the 2019-2020 season or the median of the 2016-2017 season is higher. Explain what influenced your hypothesis.

Answers will vary. My hypothesis is that the median of the 2016-2017 season is higher because most of the values for the 2019-2020 are closer together and are smaller values.

- d. Compare your predictions with your partner's predictions. Make any adjustments needed based on your conversation.
2. The table shows the list of the top 15 women goal scorers during the 2016-2017 NCAA Women's Division 1 soccer season.

Rank	Player	Goals
1	Savannah Jordan, Florida	81
2	Tabitha Tindell, FGCU	66
3	Rachel Hill, UConn	61
4	Mckenzie Meehan, Boston College	58
5	Aaliyah Lewis, Alabama St.	57
6	Tyler Lussi, Princeton	53
7	Valerie Sanderson, Memphis	52
8	Murielle Tiernan, Virginia Tech	49
9	Ashley Hatch, BYU	47
10	Ariela Lewis, Alabama St.	45
11	Emily Gingrich, Saint Joseph's	43
12	Margaret Purce, Harvard	42
12	Jaycie Johnson, Nebraska	42
14	Hayley Dowd, Boston College	39
15	Rachel Holden, North Texas	38
15	Lauren Miller, North Dakota St.	38
15	Abby Reed, DePaul	38

- a. Find the mean number of goals for the 2016-2017 soccer season.

49.9



Encourage students to analyze and respond to their partners' predictions to encourage mathematical discussion. Ensure students are using formal vocabulary, such as variability, spread, symmetry, and distribution to justify their responses.

9.8.4 Your Turn! (continued)

- b. Interpret the mean.

Answers will vary. The average goals scored for the top 15 women during the 2016-2017 NCAA Women's Division 1 soccer season was 49.9 goals.

- c. Find the median number of goals for the 2016-2017 soccer season.

47

- d. Interpret the median.

Answers will vary. The median number of goals scored for the top 15 women during the 2016-2017 NCAA Women's Division 1 soccer season was 47 goals. Half of the top 15 scored fewer than 47 goals and half scored more than 47 goals.

- e. Complete the statements.

The mean of the 2016-2017 soccer season is

- ☐ less
☒ greater

than the mean of the 2019-2020 soccer

season. The value in the ☐ 2016-2017
☐ 2019-2020 soccer

season that likely influenced the mean is 81.

The median of the 2016-2017 soccer season is

- ☐ less
☒ greater

than the median of the 2019-2020

soccer season.



For a project-based learning activity, consider allowing students to research data on a topic they are interested in that has two different sets of data to create tables and line plots for each set of data to compare.

9.8.5 Wrap-Up: Ticket Out the Door

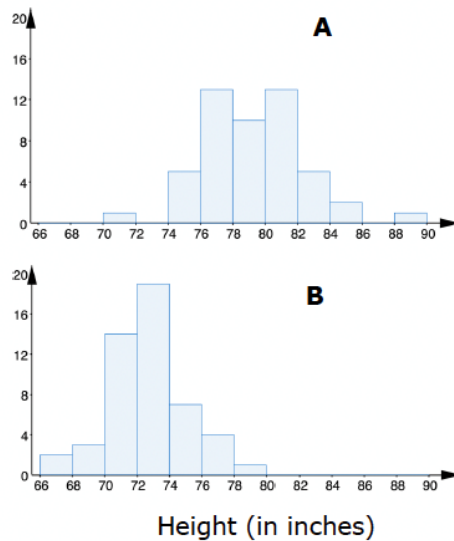
Component Overview: Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' abilities to compare distribution in data sets.



9.8.5 Wrap-Up: Ticket Out the Door

1. Professional basketball players tend to be taller than professional baseball players.

The two histograms, labeled A and B, show height distributions of 50 male professional baseball players and 50 male professional basketball players.



Data from this Ticket Out the Door may be useful to inform small groups for remediation or reteaching opportunities.

- a. Choose one the titles: "Heights of Baseball Players" or "Heights of Basketball Players" for each histogram.

Histogram	A	B
Best Title	Heights of Basketball Players	Heights of Baseball Players

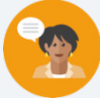
- b. Describe the spread and shape of the distribution of the heights of the basketball players.

Answers will vary. The heights of professional basketball players are symmetric, with most players between 76 and 82 inches tall. The spread of the distribution shows a range from 70 inches tall up to 90 inches tall.

9.9 Creating Box Plots

Lesson Introduction

 Benchmarks	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.		 Learning Targets	- I can find key values given a box plot within a real-world context. - I can create a box plot to represent data.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.		MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.	
 Essential Questions	- What information does the graph give you about the data? - What is an advantage of displaying the data in a graph rather than a list of values? - Why would you choose a box plot graph for this data?			
 Lesson Narrative	In this lesson, students use box plots to represent data sets. This lesson begins with students reviewing vocabulary related to data sets and extends this knowledge to representing data using box plots. Students apply what they know to create box plots to represent given data sets using the five-number summary, and the lesson culminates with a self-assessment of the lesson's learning.			
 Component Summary	9.9.1 Warm-Up (~5 min.)	Calculate the median and range for a group of data (<i>SE p. 90</i>)	9.9.4 Guided Practice (5 min.)	Create a box plot with a set of data points with assistance (<i>SE p. 93</i>)
	9.9.2 Refresher (5-10 min.) (Optional)	Label a blank box plot and review vocabulary (<i>SE p. 90</i>)	9.9.5 Try It (5-8 min.)	Create a box plot with a set of data points (<i>SE p. 93</i>)
	9.9.3 Guided Instruction (15 min.)	Complete the five-number summary for a two data sets (<i>SE p. 91</i>)	9.9.6 Your Turn! (10 min.)	With a partner, discuss the five-number summary and complete a box plot (<i>SE p. 94</i>)
	9.9.7 Wrap-Up (~5 min.)	Reflection – Rank your understanding of today's assignment (<i>SE p. 96</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Five-number summary <u>Additional Terms:</u> - Box plot - Interquartile range - Median - Range - Minimum - Maximum - First quartile - Third quartile		(Optional) Foldable/interactive notebook	N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students might confuse the percentages contained in a given interval of the boxplot. - Students may mistakenly label the median as the mean. - Students may misread the graph, especially if comparing two graphs. - Students may forget to order data before finding the five-number summary.	- Stress the percentage breakdown of the quartiles in the graph, discussing the percentage of data values above, below, and in between values. - Have students find the five-number summary before creating the box plot. - Discuss that while the box plot displays the overall shape of the distribution, individual data values and unusual features such as gaps, clusters, and outliers cannot be determined from the box plot. - If the data contains larger values, the <i>x-axis</i> does not have to start at zero. Use the range of the data to determine an appropriate scale. - The IQR represent the middle 50 percent of the data.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 9.9.3 Guided Instruction provides an opportunity to engage students in a Notice and Wonder routine to position students to consider how to represent data sets with an even and odd numbers of data points using box plots.	MA.K12.MTR.2.1: Multiple Representations Students will be working to represent box plots. Clarifications within this standard emphasize connections between concept and representation. Utilize the progression of the lesson to highlight the process by which students are moving from finding the data to representing it in a box plot.
 Differentiate Instruction	9.9.4 Guided Practice - Consider differentiating this component by ability. Use informal data gathered earlier in the lesson to group students. Allow students who demonstrate understanding to work independently, students who may benefit from discussion to work in pairs and pull a small group of students who are struggling to guide them through this component verbally. The 9.9.7 Wrap-Up requires a written explanation as part of a self-assessment. For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.	
Lesson Notes:		

9.9.1 Warm-Up: Sleep

Component Overview: Students find the median and range of a data set, two measures used to help construct box plots later in this lesson.



9.9.1 Warm-Up: Sleep

Ten sixth-grade students were asked how much sleep, in hours, they usually get on a school night. Their responses are shown.

6, 7, 6.5, 7.5, 8, 8.5, 7, 6.5, 5, 7

1. Calculate the median number of hours of sleep for the students.
7
2. Calculate the range for number of hours of sleep for the students.
3.5



Students will use their knowledge of median and range and apply it to box plots throughout this lesson. Activate prior knowledge through intentional questioning:

- What is the median of a data set? How can you arrange the data to determine the median?
- What does the median represent in this problem?

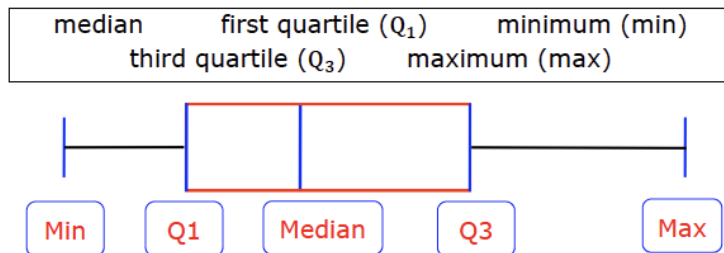
9.9.2 Refresher: Box Plots

Component Overview: Students make connections between box plots and the vocabulary used to describe them. In the previous lessons, students used the vocabulary to interpret box plots. This component is an optional refresher.



9.9.2 Refresher: Box Plots

A blank box plot is shown. Label the blue boxes below using the words found in the bank.



Consider having the students practice just finding the five-number summary before they must draw and label a box plot.

1. Complete the statements.

The **interquartile range** is a measure of spread determined by finding the distance between the first quartile and the third quartile. It measures the spread of the middle **50 percent** of the data.

The **range** is another measure of spread determined by finding the difference between the largest data value and the smallest data value.

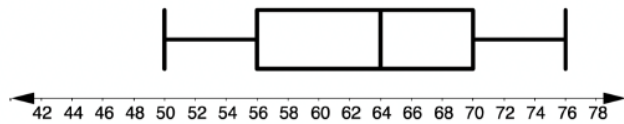
2. Complete each statement.

- a. **25%** of the data falls between the minimum and the first quartile (Q_1).
- b. **75%** of the data falls between the minimum and the third quartile (Q_3).
- c. **50%** of the data falls between the first quartile (Q_1) and the third quartile (Q_3).

9.9.2 Refresher: Box Plots (continued)

3. The box plot shows the distribution of the daily number of people who were at a large community pool in July 2018.

Number of Patrons at Community Pool



- What is the median number of people at the pool each day?
64
- What is the maximum number of people?
76
- What is the minimum number of people?
50
- Determine the value of the first quartile (Q_1) from the box plot.
56
- Determine the value of the third quartile (Q_3) from the box plot.
70



The **five-number summary** are the values of the minimum, first quartile (Q_1), median, third quartile (Q_3), and the maximum. These are the values used to create a box plot to represent a data set.



What does it mean if the median line is not in the middle of the box?



Have the students tell you in words how to find the first and third quartile.



Consider a foldable or type of interactive notebook for the five-number summary. Examples, non-examples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

9.9.3 Guided Instruction: The Five-Number Summary

Component Overview: This component combines all the aspects of finding the five-number summary. During this guided instruction, students work with a partner to apply the definitions of the five-number summary to various data sets. Students then use this information to analyze and interpret the data.



9.9.3 Guided Instruction: The Five-Number Summary

1. The ages of 20 customers who attended a special event at a local restaurant are shown in order from least to greatest.

7, 8, 9, 10, 10, 11, 12, 15, 16, 20, 20,
22, 23, 24, 28, 30, 33, 35, 38, 42

- a. Find the five-number summary of the data set.

Minimum: 7

First quartile: 10.5

Median: 20

Third quartile: 29

Maximum: 42

- b. Discuss with your partner if the data the restaurant collected is reflective of the ages of customers who eat at the restaurant.

Answers will vary. It is reflective of the special event. It is not reflective of normal traffic at this restaurant.



This component provides auditory entry points for learners through discussion. Visual entry points can be provided through encouraging students to represent data using a box plot. Connecting the model to the vocabulary will help students synthesize the vocabulary rather than simply memorizing the definitions.



Truthful data is data that is random, representative of the population, and has a large sample size.

9.9.3 Guided Instruction: The Five-Number Summary (continued)

2. The birth weights, in ounces, of golden retriever puppies born at a kennel were recorded for the past month. The data is recorded in the table.

13	14	15	15	16	16	16	16	17
17	17	17	17	17	17	18	18	18
18	18	18	18	18	19	20		



As students are working together to complete this component, encourage them to use mathematical vocabulary and language while explaining their process for finding the five-number summary with their partner. This component should help students formalize their understanding of the definitions as they continue to apply them to analyze data sets.

- a. Find the five-number summary of the data set.

Minimum: 13

First quartile: 16

Median: 17

Third quartile: 18

Maximum: 20

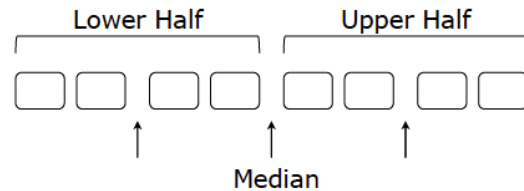
- b. Discuss with your partner why the data may not be a true representation of the birth weight of all puppies.

Answers will vary. Because this is a single breed, Golden Retriever, it does not represent all breeds. The size of the parents may have an effect on the birth weights of the puppies.

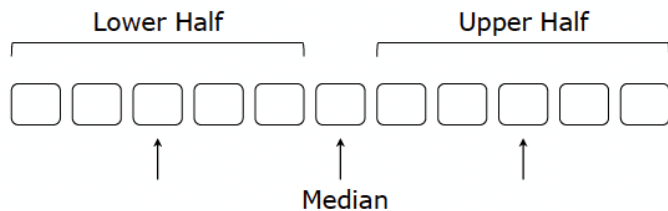
3. What differences did you notice when calculating the five-number summary for the two different data sets?
- Answers will vary. The first data set has a larger range. The second data set has more entries.*

9.9.3 Guided Instruction: The Five-Number Summary (continued)

An illustration of the five-number summary for an even data set is shown.



An illustration of the five-number summary for an odd data set is shown.



Notice and Wonder

Consider structuring this component with a Notice and Wonder routine. Students take a few moments to observe each illustration and make notes of what they notice about it and what they wonder. Then they discuss with a partner and compare their responses. After students have had a chance to share with a partner, bring the class together and record notices and wonderings on chart paper or a white board and draw attention to themes in student responses.

Students commonly divide the data incorrectly when calculating quartile values, causing their constructed box plots to be incorrect.

4. Discuss with your partner your observations and any questions you have about the illustrations.

Answers will vary. The median and first and third quartiles will not always be equal to one of the values in the data set. The quartile may fall between two values, in which case the average is found to determine the value of the quartile.

9.9.4 Guided Practice: Creating a Box Plot

Component Overview: In this Guided Practice, students follow steps to create a box plot using provided data. Use informal pieces of feedback or data by questioning students regarding prior components. Consider mobilizing some students to practice independently, some students to practice in pairs, and some students to be in a small group with the teacher.



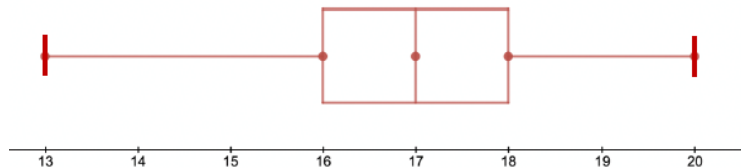
9.9.4 Guided Practice: Creating a Box Plot

- Using the data and five-number summary on the birth weight, in ounces, of the golden retriever puppies born at a kennel in the last month, create a box plot from the steps listed.

13	14	15	15	16	16	16	16	17
17	17	17	17	17	17	18	18	18
18	18	18	18	18	19	20		

Step 1: Label the axis by starting at a number slightly less than the minimum. Increase by the same interval, making sure each interval is equally spaced, until you reach (or pass) the maximum.

Birth weight, in ounces, of golden retriever puppies



Step 2: Place a dot above the axis for each value of the five-number summary.

Step 3: Draw the box from Q_1 to Q_3 with a line inside the box to represent the median.

Step 4: Connect the minimum and the maximum to each side of the box with a line.

Step 5: Include a title and a label for the axis.

- Compare your box plots with your partner. Make any revisions if necessary.



Consider differentiating this component by ability. Use informal data gathered earlier in the lesson to group students. Allow students who demonstrate understanding to work independently, students who may benefit from discussion to work in pairs and pull a small group of students who are struggling to guide them through this component verbally.

9.9.5 Try It: Creating a Box Plot

Component Overview: Students find the five-number summary of a set of data and apply it to create a box plot. Students then answer questions based on the box plot and data. Consider allowing students to work in pairs to encourage mathematical discourse as they work through this component.



9.9.5 Try It: Creating a Box Plot

1. The ages of the 21 restaurant customers who attended a special event at a local restaurant are shown.

27, 28, 39, 18, 18, 21, 32, 25, 46, 40, 40,
22, 23, 24, 28, 30, 33, 35, 38, 42, 57

- a. Determine the five-number summary for the data set.

18, 18, 21, 22, 23, 24, 25, 27, 28, 28, 30,
32, 33, 35, 38, 39, 40, 40, 42, 46, 57

Minimum: 18

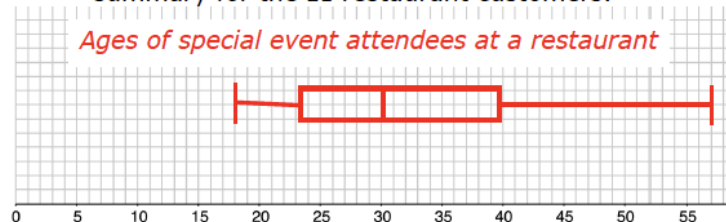
First Quartile: 23.5

Median: 30

Third Quartile: 39.5

Maximum: 57

- b. Create a box plot using the data and five-number summary for the 21 restaurant customers.



- c. Does the length of each part of the box plot need to be the same?
No
- d. Why is the actual length of each part not equal, even though they each represent 25% of the data?
Answers will vary. There are less numbers in the data set that are further from the median, causing the data to not be divided equally. The distribution of the ages of the customer will determine how the boxes are drawn. Ages closer together will skew the box one way or another.
- e. What is the interquartile range for the data?
16
- f. What is the range?
39



Students should first recognize that the data needs to be listed in ascending order to determine the five-number summary.



Circulate to ensure students are using the five-number summary to create their box plot. To deepen conceptual understanding, ask questions like:

- How can you determine where the box plot whisker should start? How can you determine where it should end?
- Will the median be in the middle of the box plot? Why or why not?
- How can you determine where the quartiles should be?

9.9.6 Your Turn!

Component Overview: Students are given the opportunity to take what was practiced in the Guided Practice and Try It components and apply their knowledge as they work with a partner. Probe students as they work using assessing and critical thinking questions to check for understanding.



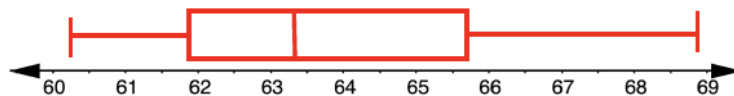
9.9.6 Your Turn!

1. Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. The five-number summary for his pass-completion percentages from 2000 to 2009 is shown.

Minimum	Maximum	Q_1	Q_3	Median
60.2	68.9	61.8	65.7	63.3

- a. Discuss with your partner why the data used to determine the five-number summary for Tom Brady's pass completion is more truthful than the data for golden retriever puppies, or the ages of the restaurant customers who attended a special event.
Answers will vary. The data is more comprehensive, it is over a ten-year period. The puppies data consisted of one month of litters and the restaurant customers consisted of one special event.
- b. Use the five-number summary to draw and label a box plot of the data set.

Tom Brady's Completion Percentages 2000-2009



- c. Determine an alternative scale that could be used to make the box plot.
I could have also plotted by the whole number.



Within this benchmark, it is the expectation to use appropriate titles, labels, scales, and units when constructing graphical representations. Circulate to ensure students are using appropriate scales and making this connection.

9.9.6 Your Turn! (continued)

- d. What scale would you use if the data values ranged from 10.3 to 48.4?

Answers will vary. By 2.

- e. Label the number line with your scale from Part D. Then, compare it with your partner.



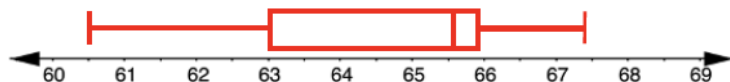
2. Tom Brady's pass-completion percentages from 2009 to 2019 are listed in the table.

Year	2009	2010	2011	2012	2013	2014
%	65.7	65.9	65.6	63.0	60.5	64.1
Year	2015	2016	2017	2018	2019	
%	64.4	67.4	66.3	65.8	60.8	

60.5, 60.8, 63.0, 64.1, 64.4, 65.6, 65.7, 65.8, 65.9, 66.3, 67.4
Minimum 60.5, First Quartile 63.0, Median 65.6, Third Quartile 65.9, Maximum 67.4.

- a. Create a box plot of Tom Brady's pass-completion percentage.

Tom Brady's Pass-Completion Percentages 2009-2019



- b. Determine the interquartile range for Tom Brady's pass-completion percentage.

2.9

- c. What is the range of pass-completion percentages?

6.9



If you see students struggling to create the box plot, provide additional scaffolding through questioning:

- *What scale makes sense for these numbers? Why?*
- *How can you determine where the first quartile should start?*
- *How can you determine the median? Where will this be represented in the box plot?*



Consider a peer review process for students to check each other's work as an added layer of support.

9.9.7 Wrap-Up: Reflection

Component Overview: Students reflect on their understanding of creating box plots. Students will need these skills when creating displays.



9.9.7 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of creating box plots. Then provide a brief explanation or evidence to support your self-assessment.

1. I can find the five-number summary of a data set.

I need help. I'm getting there. I understand
I can't get started. and have evidence.



Answers will vary.

2. I can create an axis with the appropriate scale.

I need help. I'm getting there. I understand
I can't get started. and have evidence.



Answers will vary.

3. I can find and interpret the interquartile range.

I need help. I'm getting there. I understand
I can't get started. and have evidence.



Answers will vary.

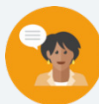


For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.

9.10 Creating Histograms

Lesson Introduction

 Benchmark	MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.		 Learning Target	I can create a histogram to represent data.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.		MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.	
 Essential Questions	- What does the skew of the graph tell you? - Why can't you give the exact value of the median using the histogram? - How can you tell the difference between a gap, an outlier, and a cluster?			
 Lesson Narrative	Students will learn how to create histograms from a given set of data. They will describe the distribution's shape and features such as outliers, gaps, or clusters. Students will use the histogram to find the range and discuss the measures of center for the data.			
 Component Summary	9.10.1 Warm-Up (~5 min.)	Notice and wonder from a graphical display (<i>SE p. 97</i>)	9.10.3 Guided Practice (10-15 min.)	Complete a frequency table and histogram (<i>SE p. 100</i>)
	9.10.2 Guided Instruction (15-20 min.)	Work with a partner to calculate the range, interval width, minimum value, first interval, and table with frequencies (<i>SE p. 97</i>)	9.10.4 Your Turn! (5-8 min.)	Independently complete a frequency table and histogram for the given data (<i>SE p. 102</i>)
	9.10.5 Wrap-Up (~2 min.)	Reflect on the lesson and anything they are still confused by (<i>SE p. 102</i>)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> - Frequency table <u>Additional Terms:</u> - Interval width - Frequency	N/A	N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse left and right skew. - Students may misread the data, using the frequencies for the range instead of the x-axis. - Students may call every gap in a histogram an outlier instead of using the term for larger gaps. - Students may attempt to give the mean or median an exact value when they can usually only approximate the interval that contains the value.	- Emphasize that the bars should touch in a histogram unless there is a gap or outlier. - Always use context when describing a distributions shape and features. You can use words like “unimodal” or “bimodal” if appropriate for the data. - The bin width is determined by width of the interval on the x-axis.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 9.10.1 Warm-Up engages the class in a Notice and Wonder routine. Encourage students to make connections between their notices and wonderings and those of their classmates to build investment in the content of the lesson.	MA.K12.MTR.4.1: Discussion and Reflection 9.10.3 Guided Practice provides an opportunity for students to reflect on the mathematical thinking of others, particularly to recognize errors and explain how to fix them. Use this as an opportunity to engage the class in partner or whole-group discussion to explain their responses and articulate their thinking.
 Differentiate Instruction	9.10.2 Guided Instruction provides multiple entry points through the use of visuals and discussion-based learning. Consider ways in which kinesthetic entry points can be provided, such as online math manipulatives.	
Lesson Notes:		

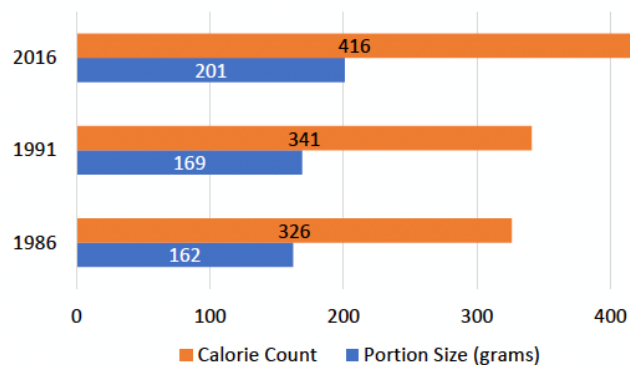
9.10.1 Warm-Up: Notice and Wonder

Component Overview: In this Warm-Up, display the image for the class to see. Then have students participate in a Notice and Wonder routine.



9.10.1 Warm-Up: Notice and Wonder

The portion size and calorie count of fast-food menu items have changed over time.



Notice and Wonder

Display the image for the class to see. Have students record what they notice about the image and what they wonder. This will engage all learners through two low-stakes questions. Bring the class back together to share their notices and wonderings. Consider recording on a T-Chart so students can make connections between their thoughts and those of their classmates.

- What are two things you notice about the graph?
Answers will vary. Portion size is less than the calorie count. The graph shows two years that are five years apart and another that is 25 years apart.
- What is one thing you wonder about the graph?
Answers will vary. Why is calorie count increasing? Why is portion size increasing?

9.10.2 Guided Instruction: Elements of Creating Histograms

Component Overview: Students collaborate to determine how to use tables to answer questions about the data presented. Students first make observations of the data before they find the necessary components for their histogram.



9.10.2 Guided Instruction: Elements of Creating Histograms

1. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. The farmer measured the lengths of 24 earthworms in one of the containers. Each length was measured in millimeters and recorded in the table.

6.5	11.2	18	19.6	20.1	23.8	23.5	25
25.7	26.2	27.5	27.5	28	29.9	32	33
41.4	42.3	48.8	52	54.4	59	60.6	77

Work with your partner to collect the pieces needed to create a histogram.

- a. The range of the data is **70.5**.

Determine the **interval width** by dividing the range and the number of intervals desired.

$$\frac{\text{Range}}{\text{Number of Intervals}} \approx \text{Interval Width}$$

- b. With your partner, discuss three different numbers of intervals you could use with your partner, and how using a different number of intervals would affect the histogram.

Answers will vary. You could use 4, 6, or 8. Using different intervals will affect the height of the histogram over certain intervals.

- c. For this histogram, use six intervals. Divide the range by **6** and round up to the nearest whole number. This gives an **interval width** of **12**.



This component is intentionally scaffolded to build on student understanding as the component progresses. Students begin by making observations about the data with a partner and then consider ways to represent the data in a frequency table and then a histogram. Ensure students understand how to create a frequency table using reasonable intervals, as this is imperative to being able to represent the data as a histogram.



Discuss how many intervals a histogram should have. The number of intervals depends on the data, but a typical histogram has five to ten intervals.



This component provides visual entry points through the use of tables and histograms. Auditory entry points are built in through partner and class discussion.

9.10.2 Guided Instruction: Elements of Creating Histograms (continued)



A **frequency table** is a table that shows how often each item, number, or range of numbers occurs in a set of data.

Use the steps below to create a frequency table to represent the data.

- d. Complete each statement.

The minimum value is 6.5, so create the first interval that starts at 6 (or less).
To get the highest value in the interval, add the interval width of 12.

The first interval is 6 – 18, the second interval is 18 – 30.

- e. With your partner, discuss if the value of 18 can be included in both intervals. Explain your conclusion.
No, each value can only be included in one interval. Otherwise, the frequencies would not be accurate.

- f. Complete the **Length (mm)** column of the table shown to indicate the intervals.

Length (mm)	Frequency
6 – 18	2
18 – 30	12
30 – 42	3
42 – 54	3
54 – 66	3
66 – 78	1

- g. With your partner, determine the frequency values and fill in the **Frequency** column of the table.
Shown in table.



Consider using visual aids, flash cards, and images to support the vocabulary and give the students time to grasp the meanings.



If you notice students struggling to determine which intervals to use, consider asking questions to further scaffold this component:

- What do all intervals have in common?
- Will this always be true? Why or why not?



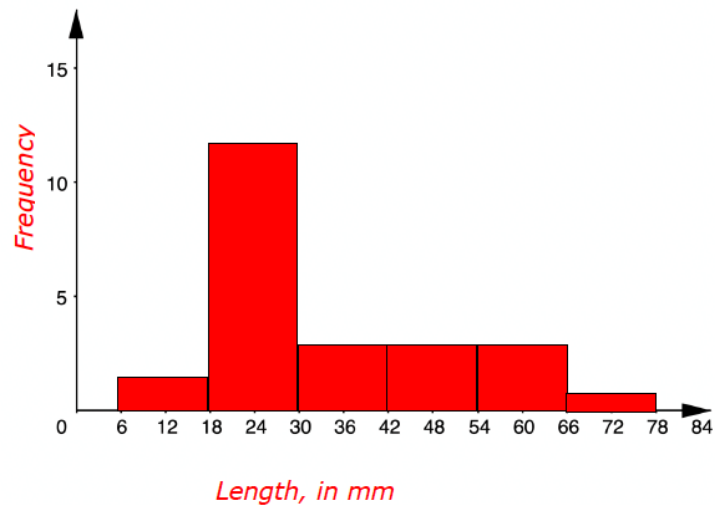
Specifically have students make connections to their responses to the first part of this question. Students who may not have made the connection to the importance of order might now better connect when given the context of a problem.

A data value of 18 should be included in the second interval, since the lower limit of an interval is included when finding frequency, but the upper limit is not.

9.10.2 Guided Instruction: Elements of Creating Histograms (continued)

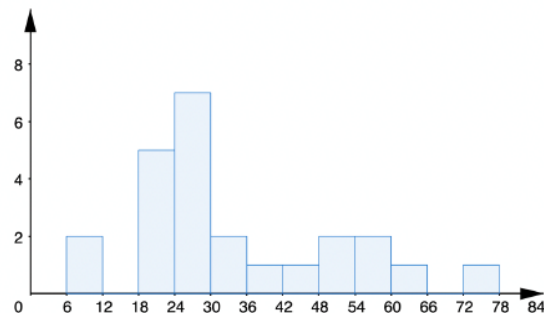
In a histogram, the bars touch. The variable measured is placed on the horizontal axis, and frequency is placed on the vertical axis.

- h. Label the horizontal and vertical axis of the grid below. Then, use the grid to create the histogram to display the lengths of earthworms in millimeters.



The sum of the frequencies in the table will always equal the number of values in the data set. Students can also check their work by adding the frequency for each bar of their histogram.

2. The same data could be displayed in this histogram.



9.10.2 Guided Instruction: Elements of Creating Histograms (continued)

- a. With your partner, discuss the differences you notice about this histogram from the one you created.

Answers will vary. The length of the interval width is 6; There are more bars in this histogram; There are haps in this histogram; The highest frequency is 7.

- b. Explain which histogram better displays the data.

Answers will vary. Histogram #2, gives a better representation of the how many worms are within a specific measurement range and a more detailed view of the shape of the distribution.

- c. Ask two classmates the reasons they listed for which histogram better displays the data. Record their choice and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Histogram Choice	My Summary of their Reason	Initials
	<i>Answers will vary.</i>	
	<i>Answers will vary.</i>	

- d. Explain whether the data has the characteristics that will result in truthful data.

Answers will vary. Yes, the data has characteristics that result in truthful data. The measurement of the worm is accurate.



Notice and Wonder

Consider implementing a Notice and Wonder routine to draw connections between the two histograms.

- What do you notice? What is the same?
- What is different?
- What does that make you wonder or think about this histogram?
- Will this always be true? Why or why not?
- How can we represent this graphically?

9.10.3 Guided Practice: Creating Histograms

Component Overview: In this Guided Practice, students create frequency tables and histograms based on provided data and then compare their interval widths with a partner. Students then analyze a provided work sample to determine if the data presented is represented correctly in a frequency table and histogram.



9.10.3 Guided Practice: Creating Histograms

1. The data shows the number of Snapchats sent by 12 seventh-grade students in one day.

99, 97, 94, 88, 84, 81, 80, 77, 71, 63, 60, 54

Complete the following to create a frequency table and a histogram to represent the number of Snapchats sent by seventh-grade students in one day.

- a. How many intervals are appropriate to use for this histogram?

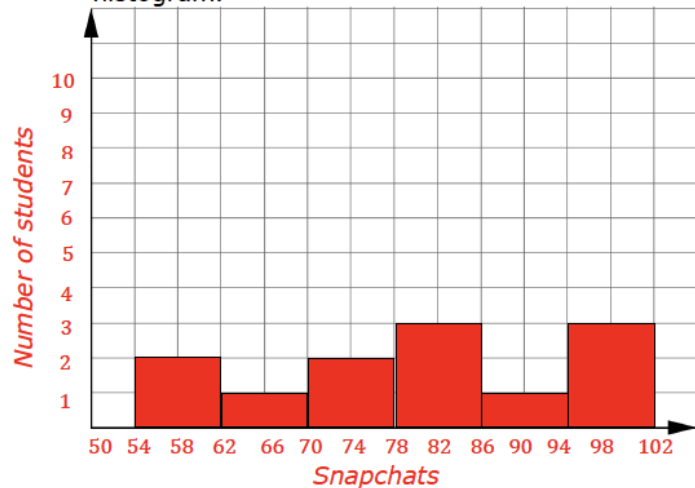
Answers will vary. 5

- b. List three possible interval widths.

Answers will vary. 5, 6, 8

- c. Using one of the possible interval widths, create the frequency table and histogram of the data. Make sure to label both the horizontal and vertical axis of the histogram.

Snapchats	Frequency
54 – 62	2
62 – 70	1
70 – 78	2
78 – 86	3
86 – 94	1
94 – 102	3



Consider building in partner discussion throughout this component to provide auditory entry points for learners. For students who may be struggling readers, it may be helpful to read the questions out loud. Consider pulling a small group to do this and allowing other students to work in pairs or small groups independently.



If you notice students struggling to create the histograms, ask scaffolding questions like:

- What intervals did you use in your frequency table?
- How can you represent these intervals on the x-axis? What scale should you use?

9.10.3 Guided Practice: Creating Histograms (continued)

- d. Find a classmate who used a different interval width and compare your histograms. Summarize your discussion on what you notice about the two different graphs.

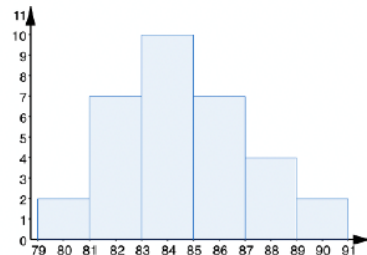
Answers will vary. They used larger interval widths, resulting in less number of intervals than I had.

2. Raisa used data she collected about the resting heart rates, in beats per minute (bpm), of 22 classmates to create a histogram. The heart rates are listed below.

89 87 85 84 90 79 83 85 86 88 84 81 88 85 83
83 86 82 83 86 82 84

She created a frequency table and used it to make a histogram. The frequency table and histogram Raisa created are shown below. Unfortunately, Raisa noticed the data she collected was not accurately displayed.

Heart Rate (bpm)	Frequency
79 – 81	2
81 – 83	7
83 – 85	10
85 – 87	7
87 – 89	4
89 – 91	2



Heart Rate (bpm)

- a. Explain what Raisa did wrong.

Raisa counted the cut scores twice.



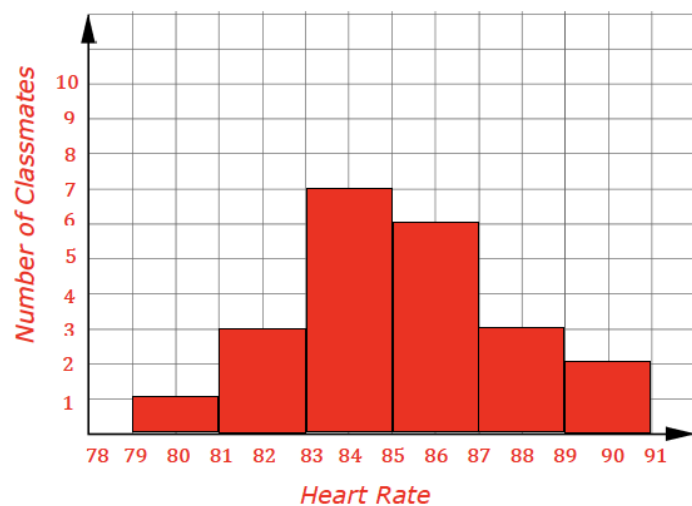
MA.K12.MTR.4.1: Discussion and Reflection

Question 2 provides an opportunity for students to reflect on the mathematical thinking of others, particularly to recognize errors and explain how to fix them. Use this as an opportunity to engage the class in partner or whole-group discussion to explain their responses and articulate their thinking.

9.10.3 Guided Practice: Creating Histograms (continued)

- b. Correct Raisa's mistake and create a frequency table and histogram that correctly displays the data she collected.

Heart Rate (bpm)	Frequency
79 – 81	1
81 – 83	3
83 – 85	7
85 – 87	6
87 – 89	3
89 – 91	2



Monitor for students who make similar mistakes to Raisa. Consider the following prompting questions:

- Why are numbers like 81 repeated in the table of values?
- Which group of frequencies would a number like 81 count for? Why?
- What does that look like on your histogram?

9.10.4 Your Turn!

Component Overview: Students work independently to create a frequency table and histogram to represent the number of chips in a bag.



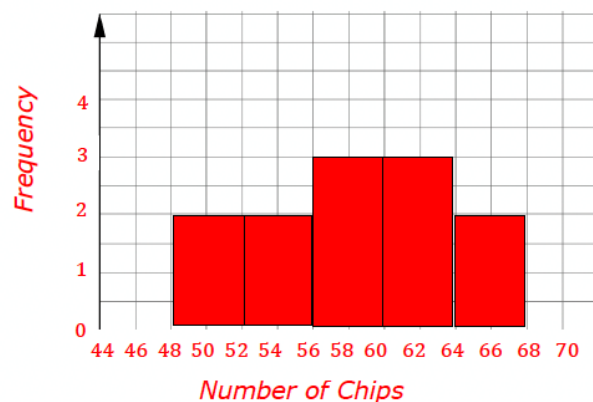
9.10.4 Your Turn!

- Sergio noticed that a bag of chips advertised that the serving size was 15 chips and that the bag contained approximately 4 servings. He counted the number of chips in a sample of 12 bags and recorded the total number of chips below.

48, 51, 52, 55, 56, 58, 59, 60, 61, 62, 65, 67

Create a frequency table and histogram for the number of chips in a bag. Answers may vary.

Number of Chips	Frequency
48 – 52	2
52 – 56	2
56 – 60	3
60 – 64	3
64 – 68	2



Students may struggle to determine an interval width that makes sense with the data. Ask guiding questions to help students determine their interval width:

- What is the range of the data? How can you use this to help determine the interval width?
- How many widths are appropriate for this data set? Why?

9.10.5 Wrap-Up: Reflection

Component Overview: Students reflect on the most confusing part of the day's lesson.



9.10.5 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on creating histograms?



Answers will vary.



Consider providing the learning targets as sentence stems for students who are unsure how to capture muddy points from the lesson.